CHAPTER

8

DNA Databases: Uses and Issues

Of any aspect of forensic DNA typing, DNA databases have arguably had the most significant impact on the criminal justice system in recent years. Serial crimes have been connected and solved. Cases without initial suspects have been brought to resolution. The innocence of unjustly incarcerated individuals has been verified when post-conviction evidence has matched another offender.

On October 13, 1998, the Federal Bureau of Investigation (FBI) officially launched its nationwide DNA database for participating law enforcement agencies. The Combined DNA Index System or CODIS involves software and hardware systems used to connect laboratories housing DNA data at the local, state, and national level in LDIS (Local DNA Index System), SDIS (State DNA Index System), or NDIS (National DNA Index System), respectively. As this chapter is being written more than 12 years later, NDIS contains over 9 million short tandem repeat (STR) profiles and links all 50 states in the United States with the capability to search criminal DNA profiles. DNA databases around the world have revolutionized the ability to use DNA profile information to link crime scene evidence to perpetrators.

These databases are effective because a majority of crimes are committed by repeat offenders. Studies have shown that more than 60% of those individuals put in prison for violent offenses and subsequently released are re-arrested for a similar offense in less than three years (McEwen & Reilly 1994, Langan & Levin 2002, Langan et al. 2003). Serious serial crimes can be connected and their perpetrators stopped through matching biological evidence between crime scenes and offenders. This chapter discusses the DNA databases being used in the United States and throughout the world to stop criminals. Issues surrounding the use and potential expansion of DNA databases are also covered.

VALUE OF DNA DATABASES

Information sharing has always been crucial to successful law enforcement. Good information can solve crimes and ultimately save lives. DNA databases have demonstrated

Advanced Topics in Forensic DNA Typing: Methodology.
DOI: 10.1016/B978-0-12-374513-2.00008-7

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their ability to serve as valuable tools in aiding law enforcement investigations (Gabriel et al. 2010). Their effectiveness continues to grow as the databases get larger. These databases can be used to locate suspects in crime cases that would otherwise never have been solved. However, with the growth and success of DNA databases, privacy concerns have been raised regarding the use and potential expansion of DNA databases as discussed later in the chapter.

An important role that DNA databases can serve is to make associations between groups of unsolved cases. Criminals do not honor the same geographical boundaries that law enforcement personnel do. Crimes committed in Florida can be linked to those committed in Virginia through an effective national DNA database.

DNA profile information must be in the database for it to be of value. For many years, tremendous sample backlogs have existed in the United States (Lovirich et al. 2004, Nelson 2010)—meaning that samples have been collected from either crime scenes or qualifying offenders but are awaiting analysis and entry into the DNA database. In order to try to alleviate this sample backlog, hundreds of millions of dollars have been poured into U.S. forensic DNA laboratories since U.S. Attorney General John Ashcroft announced the President's DNA Initiative in March 2003.

The Debbie Smith Act of 2004 (42 U.S.C 13701), which was named for a sexual assault victim whose attacker was eventually identified through a DNA database match, provides federal funding to state and local governments to perform DNA testing of backlogged samples. This funding was initially approved by Congress in 2004 and renewed in 2008. From 2005 to 2009, the U.S. government committed over $150 million per year for backlog reduction. With the renewal of the Debbie Smith Act, this funding for increasing the numbers of samples in DNA databases should continue through 2014. For more information on this initiative and results, see http://www.dna.gov.

The result of this influx of substantial funding has been the rapid expansion of the DNA database size in the United States. Yet sample backlogs still exist because the success of DNA testing has encouraged more sample submissions (Nelson 2010). In addition, expanded laws enable a wider collection of qualified offenders that translates into larger numbers of DNA samples that must be processed. Many U.S. states now have laws requiring those arrested for any felony offense to have their DNA collected and submitted to a DNA database.

The establishment of an effective DNA database requires time and full cooperation between forensic DNA laboratories, the law enforcement community, and government policy makers. An analysis of the return on this investment illustrates the worth of this work to society and especially to victims of crime (Wickenheiser 2004; see Butler 2010, *Fundamentals*, D.N.A. Box 12.1).

Definitions: Database vs. Databank

It is helpful to define several terms as we will use them in the course of this book: database, databank, and population database. A *database* is a collection of computer files containing entries of DNA profiles that can be searched to look for potential matches. In the case of essentially all forensic laboratories today, a DNA profile consists of a listing of STR genotypes produced through the process described in the previous chapters.

However, because each country may choose to use a different set of core STR markers, not all data will be compatible between countries or laboratories if there was ever a desire to search someone else's database with your unknown crime scene profile. The need for a compatible currency of data exchange is the reason that the 13 CODIS core STR loci were selected in the United States in November 1997 (see Chapter 5) almost a year before launching NDIS in October 1998.

The samples from which STR profiles are generated are usually from forensic casework or a criminal offender who has been deemed to legally qualify for entry into the database. A *databank*, as the term will be used in this book, is a collection of the actual samples—usually in the form of a blood sample or buccal swab or their DNA extracts. Most jurisdictions permit the retention of the biological specimen even after the STR typing results have been obtained and the DNA profile entered into the database. This sample retention is for quality control purposes (including hit confirmation) and enables testing of additional STRs or other genetic loci should a new technology be developed in the future. However, as will be discussed later in the chapter, sample retention is one of the first points raised by critics as inappropriate due to their fear that other genetic information will be analyzed from these samples.

Finally, information on allele frequencies from a group or groups of representative samples is included in a *population database*. Again, as a database, this refers to a collection of DNA profiles. However, this population database is not used for any kind of sample matching purposes. Rather, it is used to estimate random match probabilities based on allele frequency measurements from a group of usually 100 or more individuals selected to represent a specific group of interest. The individuals from which DNA profiles are generated for use in a population database are completely anonymous and only classified and grouped by their self-identified ethnicity. Within the CODIS software, a computer program known as PopStats performs the match probability calculations using allele frequencies from a previously typed set of samples whose STR profiles comprise the various classifications of the population database (e.g., African American, Caucasian, etc.).

Aspects of a National DNA Database

Implementing an effective national DNA database is an enormous task. A number of components must be in place before the database can be established and actually be effective. These include:

- A commitment on the part of each state (and local) government to provide samples for the DNA database—both offender and crimescene samples;
- A common set of DNA markers or standard core set so that results can be compared between all samples entered into the database;
- Standard software and computer formats so that data can be transferred between laboratories and a secure computer network to connect the various sites involved in the database (if more than one laboratory is submitting data); and
- Quality standards so that everyone can rely on results from each laboratory.

The technology of forensic DNA databases basically involves three parts: (1) collecting specimens from known criminals or other qualifying individuals as defined by law,

(2) analyzing those specimens and placing their DNA profiles in a computer database, and (3) subsequently comparing unknown or "Q" profiles obtained from crime scene evidence with the known or "K" profiles in the database. This last part often requires laboratories to work cases without suspects in order to put the crime scene DNA profiles into the database. A DNA database then enables a massive forensic unknown to offender comparison. Using these DNA databases, law enforcement agencies have been successful in identifying suspects in cases that would likely be unsolvable by any other means.

All 50 states within the U.S. have enacted legislation to establish a DNA database containing profiles from individuals convicted of specific crimes. The laws vary widely across the states concerning the scope of crimes requiring sample collection for DNA databank entry. Almost all of the states now collect samples from all felons and a growing fraction are entering DNA profiles from those arrested and accused for certain felony offenses. The trend toward broader coverage of criminal DNA databases will likely continue as these resources demonstrate their value to the criminal justice system.

ENFSI DNA Working Group Recommendations

The European Network of Forensic Science Institutes (ENFSI) DNA Working Group has produced a nice set of recommendations on DNA database management. The full document, which is updated annually following the April ENFSI DNA Working Group meeting, is available on the ENFSI website: http://www.enfsi.eu/page.php?uid=98.

The 28 recommendations from the April 2010 document are listed below (ENFSI 2010):

1. Every EU/ENFSI-country should establish a forensic DNA-database and specific legislation for its implementation and management.
2. The type of crime-related stain DNA-profiles which can be included in a DNA-database should not be restricted.
3. To increase the chance of DNA-profiles of stains to match a person, the number of persons in a DNA-database who are likely to cause matches with those stains should be as high as legally (and financially) possible.
4. Managers of national DNA-databases should establish (together with other stakeholders) criteria for the inclusion of partial DNA-profiles to obtain an acceptable balance between the minimum allowable level of evidential value (maximum random match probability) of a DNA-profile and maximum number of adventitious matches a partial DNA-profile is expected to generate.
5. DNA-profiles produced by older commercial kits should be upgraded (if possible) after a match in the National DNA-database to increase the evidential value of the match and also to fulfill the criteria for international comparison if a country wants to include DNA-profiles produced by older commercial kits in international search actions.
6. The number of loci in reference samples should be the maximum of the number of loci present in the kit(s) used for the production of the DNA-profiles of the reference samples to enhance the chance of finding relevant matches with partial DNA-profiles.
7. Labs producing DNA-profiles for a DNA-database should, as a minimum, be ISO-17025 (and/or nationally equivalent) accredited and should participate in challenging proficiency tests.

8. When DNA-profiles produced from low levels of DNA are included in a DNA-database they should be recognizable and a dedicated (near) match strategy should be used for them.
9. When a new allele is observed in a DNA-profile, its presence should be confirmed by repeated DNA-isolation, PCR, capillary electrophoresis and allele calling of the DNA-profile. Only new alleles, of which the size can be accurately determined using the internal DNA-size-standard, should be included in the DNA database.
10. Alleles from loci with chromosomal anomalies should not be included in a DNA-database as they may be caused by somatic mutations which may only occur in certain tissues/body fluids.
11. Wild cards that do not represent a designated allele should not be part of the minimum number of loci/alleles required by the Prüm-matching rules.
12. The guidelines in the document of the ISFG-working group on the analysis of mixed profiles should be used for the analysis of mixed profiles.
13. A numerical match between a reference sample and a mixed profile must always be checked against the plot of the mixed profile.
14. Mixed profiles of more than 2 persons should not systematically be included in a DNA-database because they generally will produce too many adventitious matches.
15. If the removal of a DNA-profile from the DNA-database is dependent on external information, a process should be in place to give the custodian of the DNA-database access to this information preferably by means of an automated message after an event which influences the deletion date of a DNA-profile.
16. There should be a system that can be consulted by those responsible for sampling persons to see whether a person is already present in the DNA-database.
17. The system which can be consulted by those responsible for sampling persons to see whether a person is already present in the DNA-database should be combined with a rapid biometric identification system like fingerprints to verify whether a person is already present in the DNA-database.
18. Any DNA-database should have an associated elimination DNA-database (or databases). This should include laboratory staff of all categories as well as visiting maintenance personnel. Profiles from those with access to samples (e.g., police) should also be included in addition to unidentified DNA-profiles found in negative control samples which may originate in manufacturing disposables and/or chemicals. The latter category of DNA-profiles should be shared with other ENFSI-countries.
19. The occurrence of errors in DNA-databases as a result of human mistakes associated with data entry should be avoided as much as possible by automating the allele calling and the DNA-database import process. When DNA-profiles are entered manually into the DNA-database this should be done by a process which detects typing errors, for example by double (blind) entry of data.
20. As a national DNA-database regularly is subject to attention from the public, politicians and the media, a DNA-database manager should consider establishing performance parameters and making these publicly available.
21. DNA-database managers should be aware of the possibility of adventitious matches and be able to calculate their expected numbers for the matches they report. When reporting a DNA-database match, a warning should be included indicating the factors that

increase the possibility of finding an adventitious match (size of the database, number of searches, mixed and partial profiles/random match probability, presence of family members).

22. A DNA-database match report of a crime scene related DNA-profile with a person should be informative and apart from the usual indication of the evidential value of the match (RMP) it should also contain a warning indicating the possibility of finding adventitious matches (as mentioned in recommendation 21) and its implication that the match should be considered together with other information.

23. DNA-profiles should be entered into a database in a way that guarantees their correct import. Access to the DNA-database should be limited to those persons who need to have access, by physical and organizational measures. Regular back-ups should be made, stored in a safe place, and put back at regular intervals to simulate recovery from a disaster. When DNA-profiles and their associated information are present in different systems, these systems should be regularly compared to check whether they are still properly synchronized.

24. To detect false negative matches (e.g. matches which should be found but are not found because one of the DNA-profiles contains an error) regular full DNA database searches allowing one or more mismatches should be performed. When a match between two DNA-profiles contains a mismatch in one of the loci, the original data of both DNA-profiles should be checked to see if one of the DNA-profiles contains an error.

25. Information from a National DNA-database should be combined with other types of evidence to increase the number of crimes for which a lead can be identified.

26. As automated processes reduce the possibility of human errors, they should be introduced for those processes that are straight forward.

27. From a forensic point of view the cell material of reference samples should be stored as long as their corresponding DNA-profiles.

28. Because DNA-databases have a very important but also very delicate role in society, the custodian of a DNA-database should develop tools to make objective information about the DNA-database available to politicians, the public and the media.

NATIONAL DNA DATABASES AROUND THE WORLD

A number of countries around the world have started national DNA databases (Table 8.1). The first national DNA databank, and so far the most effective (and aggressive in its application), was created in the United Kingdom in April 1995. The tremendous success of DNA in aiding crime solving has spawned extensive growth of DNA testing and formation of DNA databases around the globe.

According to a 2008 Interpol survey of 186 countries around the world, 120 countries (65%) are performing DNA profiling and 54 countries (29%) have national DNA databases (Table 8.1). Over 16 million DNA profiles existed at the time of the survey with roughly 8% of the total (1.2 million) representing crime scene evidence and the rest being from known individuals such as convicted offenders, arrestees, or volunteers. As of this 2008 survey, less than 20,000 samples had been collected worldwide from missing persons or unidentified human remains (Table 8.1).

TABLE 8.1 Summary of Global DNA Profiling and Databases.

Interpol Region		Africa	Americas	Asia & South Pacific	Europe	North Africa & Middle East	Total
Number of Countries		48	37	35	49	17	186
DNA Profiling		14	27	20	46	13	120
DNA Database		2	5	7	31	9	54
Number of Samples	# countries contributing						
Crime Scene	71	66,941	279,475	257,947	667,738	4,281	1,276,382
Convicted Offenders	44	0	6,424,724	1,052,150	287,149	25,672	7,789,695
Suspects/Arrestees	48	33,781	136,296	309,326	726,171	1,322	1,206,896
Convict or Suspect	12	–	–	–	5,990,593	–	5,990,593
Victim/Volunteer	41	21,262	587	66,141	2,133	2,250	92,373
Other	27	0	6,700	31,501	36,210	72	74,483
Missing Person	34	0	519	30	2,955	1	3,505
Unidentified Human Remains	48	0	2,285	9,615	1,588	1,521	15,009
						TOTAL	**16,448,936**

Source: Interpol DNA Unit (Interpol Global DNA Profiling 2008 Survey, Table 6.3, p. 67).

The UK National DNA Database (NDNAD)

On April 10, 1995 the world's first national DNA database, often referred to as NDNAD, was launched by the United Kingdom's Home Office (Werrett 1997). This database originally stored data from only six STR loci from the Second Generation Multiplex (SGM) consisting of FGA, TH01, VWA, D8S1179, D18S51, and D21S11. In 1999, an expansion was made to 10 STR loci (the six SGM loci plus D3S1358, D16S539, D2S1338, and D19S433) using the SGM Plus kit from Applied Biosystems. In 2009, the European Union voted to require additional STR loci in order to enable greater DNA data sharing across European countries (see Chapter 5). Plans for a future Pan-European database will likely expand the number of core loci used in the UK to 15 STRs.

In its first five years (1995 to 2000), more than 500,000 DNA profiles were entered into the database and more than 50,000 criminal investigations were aided. As of 2010, the NDNAD contains more than four million profiles and regularly aids UK law enforcement personnel in resolving thousands of crimes each year. With around 50 million people in the UK, their DNA database of greater than four million profiles represents the highest proportion of its population.

A survey in 2004 (Asplen 2004) showed at the time that the NDNAD delivered over 1700 crime scene-to-crime scene or suspect-to-crime scene hits per week! The UK government invested more than £182 million into NDNAD in the first 10 years of its existence, which equates to approximately £3 ($5) per citizen invested in DNA databasing.

Since April 2007, the UK NDNAD has been the responsibility of the National Policing Improvement Agency (NPIA), whose role it is to run database operations and to maintain integrity of the data. A National DNA Database Strategy Board governs and provides oversight of the operation of the NDNAD.

The UK database has come under intense scrutiny since a December 2008 ruling by the European Court of Human Rights. In the case of *S. and Marper v. The United Kingdom*, it was decided that retaining DNA profiles of individuals who were arrested but not convicted was a violation of the right to privacy under the European Convention of Human Rights (see ruling at http://www.webcitation.org/5g6FzdBr4). In May 2009, the UK government responded that it would continue holding DNA profiles of convicted individuals for an indefinite period of time but that adults arrested but not convicted would have their DNA profiles deleted after a period of 6 or 12 years depending on the type of crime for which the individual was arrested. Arrested juveniles can have their profiles removed when they turn 18 if they are not convicted of a crime.

Despite the recent challenges and scrutiny, the NDNAD has been a model for many of the other DNA databases developed around the world. The success of this law enforcement tool has led to legislation in many other countries enabling the development and growth of national DNA databases.

Other National DNA Databases Around the World

National DNA databases are being used in many countries around the world, and a number of other nations are in the early stages of building their own DNA databases. The same STR markers are being used in many instances. For example, there are eight STR loci (FGA, TH01, VWA, D3S1358, D8S1179, D16S539, D18S51, and D21S11) that overlap between European and United States DNA database collection efforts. This fact will permit international collaborations on cases that warrant them although having more loci in common would be beneficial as the various DNA databases continue to grow in size. Some countries, such as Germany, use STR markers (e.g., the highly polymorphic STR locus SE33) that are not as widely applied in other national DNA databases. Therefore, information from additional genetic markers like SE33 cannot be used in linking potential cross-border crime since it would not be available in data sets from both countries.

A minimal standard set of loci known as the European Standard Set (ESS) have been adopted by the European community. The original ESS included the seven STR loci FGA, TH01, VWA, D3S1358, D8S1179, D18S51, and D21S11. As noted in Chapter 5, the European Union voted in November 2009 to adopt five additional STR loci bringing the required extended ESS to 12 STRs: FGA, TH01, VWA, D3S1358, D8S1179, D18S51, D21S11, D12S391, D1S1656, D2S441, D10S1248, and D22S1045. In addition, D16S539, D2S1338, and D19S433 are often present in commercial multiplex STR kits enabling 15 common STRs to be genotyped.

Canada, New Zealand, Australia, Japan, and a number of countries in Europe besides the United Kingdom have developed successful DNA databases. The European Network

of Forensic Science Institutes (ENFSI) provides periodic surveys and information on DNA databases and database laws in Europe on their website (see http://www.enfsi.eu). Each country has different laws regarding reasons for obtaining a DNA profile, when a profile would be expunged from the database, whether or not a DNA sample will be stored following analysis, and which STR loci are included. Most countries within the European Union are standardizing on use of the 15 STRs mentioned above, which will enable fruitful collaboration of criminal DNA information in the future. China and India, the two nations with the largest populations, have also begun efforts to produce DNA databases. As of late 2010, China reportedly has over five million DNA profiles in their national DNA database being produced by a network of more than 200 laboratories.

For international cross-border DNA comparisons, Interpol has adopted the ESS along with the sex-typing marker amelogenin as additional optional information. For more information on core STR loci used by various countries, see http://www.cstl.nist.gov/biotech/strbase/coreSTRs.htm.

International Data Comparison

Several approaches have been taken to permit examination of data across international borders. These include (1) having a network of individual DNA databases that can be queried by all approved personnel, (2) sending a query profile (typically a crime scene profile from an unknown source) to another country's DNA database administrator and requesting a search against the country's index of crime scene profiles and/or index of offender profiles, and (3) some kind of combination of either #1 or #2. Differences in legislation between countries prevent DNA databases from being unified—instead data is exchanged between countries based on agreements made and legislation authorizing such communication.

Interpol has established a platform for exchange of data called the "Interpol Gateway." In May 2005, seven European countries signed what is known as the Prüm treaty to facilitate DNA profile, fingerprint, and vehicle data exchange. Two years later, the European Union adopted the Prüm treaty requiring all member countries to exchange DNA profile information. Many countries use the CODIS software (see next section), which could help unify and standardize data storage and exchange formats.

An ISO (International Standards Organization, see Chapter 7) documentary standard for DNA data exchange is in development to aid forensic and biometrics applications. "ISO/IEC 19794 Biometric Data Interchange Formats, Part 14 – DNA Data" is scheduled to be released in May 2012. The intent of this ISO standard is to establish minimum criteria for the structure of data generated by instruments and stored in DNA databases. With consistent inputs and outputs, information sharing is possible where and when desired. Thus, international DNA data exchange would be enabled with a universal language for information content and connectivity.

THE U.S. NATIONAL DNA DATABASE

The United States Congress authorized the FBI Laboratory to establish and oversee a U.S. national DNA database with the DNA Identification Act of 1994 (Public Law 103–322). However, the FBI had started a pilot project several years earlier with 14 state and local crime laboratories to see how effective a DNA database could be with the initial concepts of

the CODIS system outlined at a meeting in 1990 (Baechtel et al. 1991). For the initial CODIS pilot project, DNA labs included the Arizona Department of Public Safety, the California Department of Justice, the Florida Department of Law Enforcement, the Minnesota Bureau of Criminal Apprehension, the Virginia Division of Forensic Science, the Washington State Patrol Crime Lab, Broward County Sheriff's Crime Lab (Ft. Lauderdale, Florida), Metro Dade Police Department Crime Lab (Miami, Florida), Orange County (California) Sheriff's Crime Lab, and Washoe County Sheriff's Office Crime Lab (Reno, Nevada).

It took several years to gather enough DNA profiles from convicted offenders to reach the critical mass necessary to obtain matches for crime scene evidence. During the 1990s, the number of samples in CODIS grew to several hundred thousand. In addition, the number of laboratories submitting data increased.

The original 1994 law also established a DNA Advisory Board, which served from 1995 to 2000, with the mandate to create quality assurance standards (see Chapter 7) to ensure the reliability of data going into the U.S. national DNA database. Since 2000, the DNA Advisory Board designated the Scientific Working Group on DNA Analysis Methods (SWGDAM) with the responsibility for recommending revisions of the Quality Assurance Standards (QAS) to the FBI Director. The most recent (as of the time this book was completed in early 2011) QAS versions for casework and database laboratories went into effect on July 1, 2009.

When NDIS was launched in October 1998, only nine states participated. The number of NDIS-participating states grew as state laws were passed that permitted DNA data collection. In addition, states had to receive the necessary computer hardware and sign an NDIS memorandum of understanding (MOU) in which they agreed to abide by NDIS Procedures before they could participate. Since July 2004, all 50 states within the U.S. participate in NDIS as do the FBI Laboratory, the U.S. Army Criminal Investigation Laboratory (USACIL), and the Bureau of Alcohol, Tobacco, and Firearms (ATF) Laboratory on the federal level.

The National DNA Index System (NDIS) manages nationwide information in a single repository maintained by the FBI Laboratory. Participating states submit their DNA profiles in order to have searches performed on a national level. The role of NDIS is to search casework and offender indices, manage candidate matches, and return results of matches to the local and/or state level.

Local, State, and National Levels

The U.S. national DNA database is composed of three tiers: local, state, and national (Figure 8.1). All three levels contain the convicted offender and casework indexes and the population data file. The original CODIS software was configurable to support any RFLP (restriction fragment length polymorphism) or PCR DNA markers although as of 2000 only STR data have been added. At the local level, or Local DNA Index System (LDIS), DNA analysts can input forensic DNA profiles and search for matches with local cases. Forensic DNA records that originate at the local level can be "uploaded" or transmitted to the state and national levels.

Each participating state within the U.S. has a single laboratory that functions as the State DNA Index System (SDIS) to manage information at the state level. SDIS enables the exchange and comparison of DNA profiles within a state and is usually operated by the agency responsible for maintaining a state's offender DNA database program.

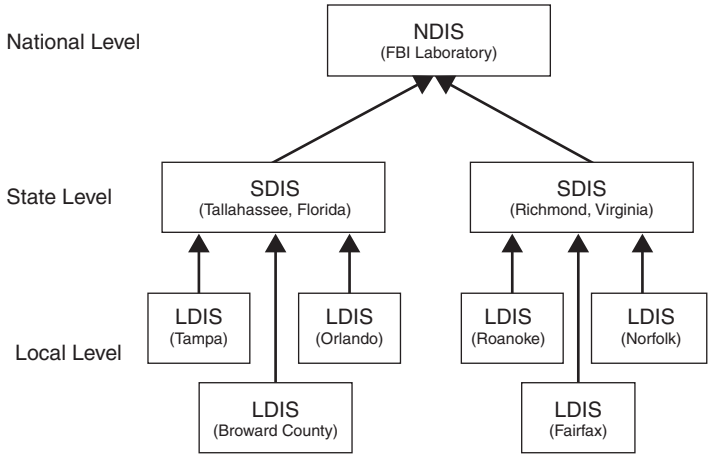


FIGURE 8.1 Schematic of the three tiers in the Combined DNA Index System (CODIS). DNA profile information begins at the local level, or Local DNA Index System (LDIS), and then can be uploaded to the state level, or State DNA Index System (SDIS), and finally to the national level, or National DNA Index System (NDIS). Each local or state laboratory maintains its portion of CODIS while the FBI Laboratory maintains the national portion (NDIS).

Combined DNA Index System (CODIS)

As noted previously, CODIS stands for the Combined DNA Index System and represents the software used to connect law enforcement laboratories housing U.S. DNA data at the local, state, and national level in LDIS, SDIS, and NDIS, respectively. These U.S. sites are all networked together on the CJIS WAN (Criminal Justice Information Systems Wide Area Network), a stand-alone law enforcement computer network that operates in a similar fashion to the Internet. The software is the same at all sites with various configurations that permit different levels of access (LDIS, SDIS, or NDIS). Software versions are updated periodically and provided to all CODIS laboratories by the FBI.

As of November 2010, CODIS software was installed in 190 U.S. laboratories representing all 50 states as well as the FBI Laboratory, USACIL, District of Columbia (Washington, DC), and Puerto Rico. This software enables NDIS-participating laboratories to submit DNA profiles for the 13 CODIS core STR loci to the U.S. national DNA database. Of the 190 sites where CODIS software is installed in the United States, 136 are LDIS and 54 are SDIS; the FBI is the only NDIS site.

The FBI has also provided a stand-alone version of the CODIS software to 68 government law enforcement laboratories in 34 different countries to aid their DNA database work. These countries include Argentina, Australia, Belgium, Bosnia, Botswana, Brazil, Canada, Cayman Islands, Chile, Columbia, Croatia, Czech Republic, Denmark, Estonia, Finland, France, Greece, Hong Kong, Hungary, Iceland, Israel, Italy, Latvia, Lithuania, Netherlands, Norway, Poland, Portugal, Singapore, Slovakia, South Korea, Spain, Sweden, and Switzerland. *It is important to note that the DNA databases in these countries use the CODIS software for their own DNA database initiatives and are not connected to the national, state, or local DNA databases in the United States.*

Each year, usually in October or November, the FBI Laboratory sponsors a National CODIS Conference to inform CODIS users of relevant issues and updates to the software and its use. Information on this conference is available at http://www.fbi.gov/about-us/lab/codis/.

CODIS Software

The CODIS software has been developed over the past two decades under multiple U.S. government contracts originally with SAIC (Scientific Applications International Corporation) and now with Unisys Corporation along with several subcontractors. A program manager within the FBI Laboratory manages the overall software development, which has cost tens of millions of dollars and taken many years to develop.

The CODIS software consists of four main subprograms: *Specimen Manager* handles the DNA profiles, *Match Manager* handles the candidate matches, *Autosearcher* conducts the database searches, and *PopStats* enables DNA profile probability calculations from population data.

CJIS WAN: The Computer Network of CODIS

Public crime laboratories in the United States are connected via the FBI's Criminal Justice Information Services Wide Area Network (CJIS WAN) through T1 lines capable of transmitting 1.5 megabytes of information per second. CJIS WAN provides Internet-like connectivity but without the security risk. This network is an intranet with access granted only to participating laboratories.

Each state pays for their end of the system. The computer equipment for a state system costs around $15,000 to $25,000. The FBI Laboratory provides the CODIS software and maintains the equipment for the national system. Each state (SDIS) laboratory signs an MOU with the FBI. LDIS labs must then adhere to the state signed MOU. CODIS users agree to adhere to FBI-issued quality assurance standards (and accreditation by a recognized accreditation source) and to submit to NDIS assessments.

Information Storage Categories or Indices

Data stored in the U.S. national DNA database are maintained in one of several categories, termed "indexes" (or "indices"). The convicted offender index and the forensic index are the two largest groups of sample data. The *convicted offender index* (sometimes shortened here to *offender index*) contains DNA profiles from individuals convicted of crimes such as sexual assault, murder, burglary, etc. On the other hand, the *forensic index* DNA profiles typically come from crime scene evidence without a suspect. However, forensic unknowns from solved crimes may be retained in the forensic index in order to help link serial crimes and aid the investigation of an unsolved case. In a sense, the forensic index provides "question" samples in the Q-K comparison while the offender index serves as the "known" sample.

With the passage of federal and state laws enabling DNA to be collected from individuals who are arrested for crimes (prior to conviction), an *arrestee index* was established a few years ago. When state laws permit collection of DNA samples at the point of arrest or

probable cause determination, these DNA profiles are housed in the arrestee index. Only states authorized to collect and database DNA samples from arrestees may upload these DNA profiles to NDIS for searching at the national level.

A national missing persons program (see Chapter 9) is supported with several separate indices in NDIS. DNA profiles from missing persons, unidentified human remains, and biological relatives of missing persons can be stored and searched against one another to try to find a direct or kinship match.

Numbers of DNA Profiles Stored

When NDIS was first activated in October 1998, there were 119,000 offender profiles and 5000 forensic casework profiles from nine states. By December 1999, 21 states and the FBI had input 211,673 offender profiles and 11,112 forensic profiles. While many of the original DNA profiles were from RFLP markers, forensic DNA laboratories in the United States converted completely to the 13 core STR loci shortly after the turn of the century. Presumably all samples for the foreseeable future will be typed with these STRs or an expanded set of loci including all or most of the original 13 core loci. At the end of 2010, the total number of offender STR profiles stood at over 9.2 million with almost 352,000 forensic profiles present in NDIS. Table 8.2 shows the growth in the number of samples at NDIS in offender, forensic, and arrestee indices from 2000 to 2010. Arrestee data have only been allowed at NDIS since 2006.

TABLE 8.2 Growth in Numbers of DNA Profiles Present in Various NDIS Indices (Cumulative Totals by Year).

Year Ending Dec 31	Forensic	Convicted Offender	Arrestee	Total Offender*
2000	21,625	441,181	–	441,181
2001	27,897	750,929	–	750,929
2002	46,177	1,247,163	–	1,247,163
2003	70,931	1,493,536	–	1,493,536
2004	93,956	2,038,514	–	2,038,514
2005	126,315	2,826,505	–	2,826,505
2006	160,582	3,977,433	54,313	4,031,748
2007	203,401	5,287,505	85,072	5,372,773
2008	248,943	6,398,874	140,719	6,539,919
2009	298,369	7,389,917	351,926	7,743,329
2010	351,951	8,559,841	668,849	9,233,554

Source: CODIS brochure available at http://www.fbi.gov/about-us/lab/codis/codis_brochure and FBI Laboratory's CODIS Unit.

**The total offender column sums the convicted offender and arrestee data and also includes a relatively small number of profiles present in a Legal Index and a Detainee Index.*

An examination of the numbers of samples in these various categories over the past several years shows several trends. First, there has been tremendous growth in the number of DNA profiles submitted to NDIS. This is due to increased funding and expansion of DNA database laws. Second, there is a much smaller number of forensic samples being submitted. The primary reason for the smaller number of crime scene samples is that it is much harder to process crime scene samples than single-source DNA samples of high quality and quantity from offenders and arrestees. Convicted-offender and arrestee samples of high quality and quantity are also in the same format (i.e., liquid blood or DNA extracted from buccal swabs), which improves the ability to automate the DNA typing process. On the other hand, forensic cases can involve the examination of a dozen or more pieces of biological evidence from a variety of formats (e.g., semen stains, bloodstains, etc.), which makes them much more challenging to process, especially if DNA degradation is found or mixtures are present.

What Information is Stored

Since they were selected in November 1997, the 13 CODIS core STR loci (see Chapter 5) have been required for data entry into the national level of the U.S. DNA database. These 13 STR markers provide a random match probability (assuming unrelated individuals) of approximately 1 in 100 trillion.

While a complete DNA profile for the 13 core STR loci is required for the convicted offender index, a minimum of 10 of the 13 CODIS STR are needed for forensic casework index information before uploading to NDIS. Fewer core STR loci were originally acceptable at the state and local levels. The lower number of loci needed for casework DNA profiles comes from recognition that degraded DNA samples obtained from forensic cases may not yield results at every marker (see Chapter 10). Effort is also made to avoid putting mixtures into the database.

CODIS is not a criminal history information database but rather a system of pointers that provides only the information necessary for making Q-K comparisons and matches. CODIS information on each DNA sample includes four pieces of information: (1) an identifier of the submitting agency, (2) the specimen ID, (3) the STR profile itself, and (4) an identifier of the analyst responsible for the profile. As can be seen with the example in Table 8.3, there are no names associated with the STR profile stored in the CODIS software indices. No personal information, criminal history information, or case-related information is contained within CODIS.

As noted in Chapter 5, the 13 CODIS STR loci are not from gene coding regions (i.e., exons) nor are any of them trinucleotides, which can be prone to expansions that cause genetic defects. An STR profile is simply a string of numbers that provides a unique genetic identifier to a tested sample. While there has occasionally been some debate in the literature regarding potential linkage of human identity testing markers to genetic disease states, this is really a non-issue (D.N.A. Box 8.1).

Occasionally there may be a desire to require new loci as a DNA database grows (see below & Chapter 5). Because of the impact that this change may have on previous information in the database, time is typically required in order to make the change. Since the Justice for All Act of 2004, the U.S. Congress must be informed by the FBI at least six months prior

TABLE 8.3 Example of the STR Profile Information Stored in the CODIS DNA Database for a Single Sample. Note That There Is No Personal Information That Can Be Used to Link an Individual to Their DNA Profile. The Two Alleles for Each STR Marker Are Placed in Separate Columns Labeled Allele 1 and Allele 2. For Markers with Homozygous Results, Both Allele 1 and Allele 2 Are the Same (e.g., CSF1PO). The Information in the "Sample ID" Field Can Be Related to a Known Individual Only by the Originating Forensic DNA Laboratory. The Date and Time Stamp Are Included to Illustrate When the DNA Profile Was Uploaded to SDIS.

Agency ID	**Sample ID**	**Analyst ID**	**Category**	
VADFS-N	1999082605	JMB	Convicted Offender	
Marker	**Allele 1**	**Allele 2**	**Date**	**Time**
AMEL	X	Y	15-FEB-2000	17:38:30
CSF1PO	10	10	15-FEB-2000	17:38:30
D13S317	11	14	15-FEB-2000	17:38:30
D16S539	9	11	15-FEB-2000	17:38:30
D18S51	14	16	15-FEB-2000	17:38:30
D21S11	28	30	15-FEB-2000	17:38:30
D3S1358	16	17	15-FEB-2000	17:38:30
D5S818	12	13	15-FEB-2000	17:38:30
D7S820	9	9	15-FEB-2000	17:38:30
D8S1179	12	14	15-FEB-2000	17:38:30
FGA	21	22	15-FEB-2000	17:38:30
TH01	6	6	15-FEB-2000	17:38:30
TPOX	8	8	15-FEB-2000	17:38:30
VWA	17	18	15-FEB-2000	17:38:30

to any new loci being added to NDIS. One example of a recent change is that the sex-typing marker amelogenin is required for DNA profiles coming from missing persons samples, relatives of missing persons, and unidentified human remains.

How Data Quality is Maintained

The old adage of "garbage in, garbage out" applies with any database containing information that will be probed regularly. If the DNA profiles entered into a DNA database are not accurate, then they will be of little value for making a meaningful match. The high quality of data going into a DNA database is ensured by requiring laboratories to follow quality assurance standards, to submit to audits of their procedures, and by conducting regular proficiency tests of analysts (see Chapter 7).

Expert systems enable automated allele calling. When combined with laboratory information management systems (see Chapter 17), which provide electronic import of data into

D.N.A. BOX 8.1

STR MARKERS USED FOR HUMAN IDENTITY TESTING DO NOT PREDICT DISEASE

An STR profile is simply a string of numbers that provides a unique genetic identifier to a tested sample (see Table 8.3). Yet because this information ultimately may be linked back to an individual, privacy concerns have been raised as to whether or not predisposition to a genetic disease can be ascertained from the presence of a particular STR allele.

In the fall of 2007, there was a brief debate in the legal literature between Simon Cole (University of California-Irvine) and David Kaye (at that time from Arizona State University) over whether or not STR markers used in forensic DNA typing could be considered "junk DNA" and free of predicting disease or physical traits (Cole 2007, Kaye 2007). Portions of this debate arose because of a misunderstanding by Simon Cole over some of the things I had written in a review article on STR markers (Butler 2006).

As I stated in the 2006 review article, it is important to keep in mind that even though medical genetic researchers claim to have shown linkage between a particular disease gene and a core STR marker, these types of findings are often tentative and should not prevent the continued use of the STR locus in question (Butler 2006). In fact, many times these linkage "findings" can later be proven false with further studies. In the past, STR markers have been used by geneticists to follow genetic disease genes through specific alleles being passed within family pedigrees. However, this use of STRs for family linkage studies is different than associations of specific alleles in a general population with a disease state. Colin Kimpton and coworkers from the European DNA Profiling Group (EDNAP) recognized early on in the application of STRs for human identity testing that "it is likely that many or possibly most STRs will eventually be shown to be useful in following a genetic disease or other genetic trait *within a family* and therefore this possibility must be recognized at the outset of the use of such systems" (Kimpton et al. 1995; emphasis added). Family pedigree studies that track a few specific loci and alleles are different than equating a specific allele in the population with some kind of phenotypic correlation.

In 2005, an infrequently used X-chromosome STR marker named HumARA was removed from future consideration in human identity testing (Szibor et al. 2005) since it is located in an exon. Some of the longer CAG repeat alleles with HumARA have been shown to be the cause of a genetic disease, which is why this STR locus was removed from use. All of the 23 commonly used STR markers described throughout this book and present in current commercial STR kits are located in between genes ("junk DNA" regions) or in introns. Thus, by definition they are non-coding. Moreover, the relatively high mutation rate of STRs means that even if any linkage existed at one time between a specific allele and a genetic disease state, this linkage would likely not last beyond a few generations before mutation altered the allele length and effectively broke any linkage of an allele or genotype state to that specific phenotype state.

Sources:

Butler, J.M. (2006). Genetics and genomics of core STR loci used in human identity testing. Journal of Forensic Sciences, *51, 253–265.*

D.N.A. BOX 8.1 (*cont'd*)

Cole, S.A. (2007). Is the 'junk' DNA designation bunk? Northwestern University Law Review Colloquy, *102, 54–63. Available at http://www.law.northwestern.edu/lawreview/colloquy/2007/23/.*

Kaye, D.H. (2007). Please, let's bury the junk: the CODIS loci and the revelation of private information. Northwestern University Law Review Colloquy, *102, 70–81. Available at http://www.law.northwestern.edu/lawreview/colloquy/2007/25/.*

Kimpton, C.P., et al. (1995). Report on the second EDNAP collaborative STR exercise. Forensic Science International, *71, 137–152.*

Szibor, R., et al. (2005). Letter to the editor: the HumARA genotype is linked to spinal and bulbar muscular dystrophy and some further disease risks and should no longer be used as a DNA marker for forensic purposes. International Journal of Legal Medicine, *119, 179–180.*

the database, human intervention and error are reduced. When manual data entry is necessary, a "double-blind" approach is used, where the same DNA profile must be entered twice without being able to see the first entry (as in ENFSI 2010, recommendation #19). The software confirms that the two entries are the same before uploading the sample profile to the database.

In order for a state to have its DNA profiles included in the national DNA index system, a memorandum of understanding must be signed whereby the state DNA laboratories agree to adhere to the FBI-issued Quality Assurance Standards (QAS) and undergo regular audits and assessments to evaluate compliance of NDIS-participating laboratories to the forensic and databasing QAS in accordance with federal law. The original forensic QAS were adopted in October 1998 (April 1999 for convicted offender QAS). Revised QAS were implemented in July 2009 for both forensic and DNA databasing laboratories.

Audits are conducted annually for each NDIS-participating laboratory, and DNA analysts undergo semi-annual proficiency tests. Failure to pass this audit can result in a laboratory being disconnected from NDIS. Likewise, proficiency tests must be successfully completed in order for an analyst to continue to submit data to LDIS or SDIS for subsequent inclusion in NDIS.

Who Inputs Data

For a criminal DNA database to be successful, convicted offender DNA samples must be tested in addition to crimescene material from cases in which there is no suspect. Because the demand for DNA testing often surpasses the ability of public forensic laboratories to perform the tests, private contract laboratories have been used to reduce the sample backlogs for convicted offender as well as for some forensic casework samples. In the United States, much of this work is being performed with federal government financial assistance through grant programs administered by the National Institute of Justice (Nelson 2010).

Pursuant to federal law, as of 2011 only government law enforcement forensic DNA laboratories who comply with the QAS have access to CODIS. Therefore, while data may be generated by a private laboratory through outsourcing from a public forensic laboratory, the

TABLE 8.4 List of NDIS-Approved Expert Systems for Single-Source DNA Data Review (as of December 2010). Information Courtesy of Dr. Douglas Hares, FBI Laboratory CODIS Unit.

Expert System and Version(s) [Manufacturer]	Instrument Platform(s)	Kit(s)
GeneMapper*ID-X* v1.0 [Applied Biosystems]	AB 3100 (data collection v2.0)	Profiler Plus and COfiler
GeneMapper*ID-X* v1.0.1 [Applied Biosystems]	AB 3100 (data collection v2.0)	Profiler Plus and COfiler
GeneMapper*ID-X* v1.0.1 [Applied Biosystems]	AB 3130xl (data collection v3.0)	PowerPlex 16
GeneMapper*ID-X* v1.0.1 [Applied Biosystems]	AB 3130xl (data collection v3.0)	Identifiler
GeneMapper*ID-X* v1.0.1 [Applied Biosystems]	AB 3730 (data collection 3.0)	Identifiler
GeneMapper*ID-X* v1.1.1 [Applied Biosystems]	AB 3130xl (data collection v3.0)	Identifiler
GeneMapper*ID* v3.2.1* [Applied Biosystems]	AB 3130xl (data collection v3.0)	Identifiler
GeneMapper*ID* v3.2.1 [Applied Biosystems]	AB3130xl (data collection 3.0)	PowerPlex 16
i-Cubed v.4.0.2 using GeneMapper*ID* v3.2 [FSS/Promega and Applied Biosystems]	AB 3700 (data collection v3.1.1)	Identifiler
i-Cubed v.4.1.3 using GeneMapper*ID* v3.2 [FSS/Promega and Applied Biosystems]	AB 3130xl (data collection v3.0)	Identifiler
i-Cubed v.4.2.1 using GeneMapper*ID* v3.2 [FSS/Promega and Applied Biosystems]	AB 3130xl (data collection v3.0)	PowerPlex 16
i-Cubed v4.2.2 using GeneMapper*ID* v3.2.1 [FSS/Promega and Applied Biosystems]	AB 3130xl (data collection v3.0)	Identifiler
TrueAllele v2.7.348 [Cybergenetics]	AB 3100 (data collection v1.1)	Profiler Plus and COfiler
TrueAllele v2.9 [Cybergenetics]	AB 3100 (data collection v1.1)	Profiler Plus and COfiler
TrueAllele v2.9 [Cybergenetics]	AB 3130xl (data collection v3.0)	Identifiler
TrueAllele v2.9 [Cybergenetics]	AB 3130xl (data collection v3.0)	Profiler Plus and COfiler

**Originally approved as Genemapper ID-X 3.2, an update was provided by the vendor, retested by the submitting agency, and approved by the FBI.*

STR typing data must be evaluated by the public laboratory who assumes ownership of the data. Only then, after the quality of the STR typing data is confirmed, can it be uploaded by the public laboratory to LDIS, SDIS, and ultimately NDIS.

Over the past decade, contract laboratories have collectively generated millions of STR profiles and directly aided the rapid growth of the U.S. national DNA database. During this same time, with federal grant programs for capacity building, many government forensic laboratories have been working to build up their own capacity in order to be able to do all of their sample testing in-house. Improved automation through robotic sample handling and multi-capillary capillary electrophoresis (CE) systems have simplified the process of DNA typing and aided sample throughput. Several expert systems have been approved to review single-source DNA databasing profiles (Table 8.4).

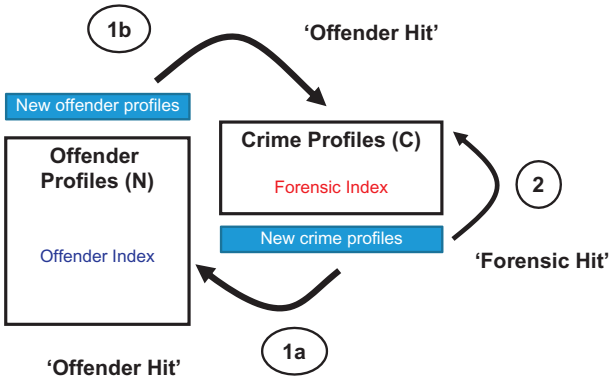


FIGURE 8.2 Primary searches conducted each week with the offender and forensic indices of the U.S. national DNA database. (1a) New crime profiles being added to the forensic index are searched against the entire offender index; (1b) new offender profiles are searched against the entire forensic index; and (2) new crime profiles are searched against the entire forensic index to help connect serial crimes. The arrestee index is treated like the offender index for these searches. A match between DNA profiles with searches 1a or 1b is termed an "offender hit" while a search 2 match results in a "forensic hit."

What Searches Are Conducted to Obtain Matches

Figure 8.2 illustrates the primary searches conducted with the two largest indices of the U.S. national DNA database and the types of "hits" or matches that can occur. An *offender hit* is produced when a match occurs between a DNA profile on the convicted-offender or arrestee index against the crime scene DNA profiles present in the forensic index during a search of offender or arrestee DNA profiles. This type of hit is sometimes referred to as a case-to-offender hit.

A *forensic hit* results from searching DNA profiles in the forensic index against other crime scene DNA profiles—essentially an effective method to look for crimes committed by a serial offender. This type of hit is sometimes referred to as a case-to-case hit.

How Success of the U.S. DNA Database Is Measured

The purpose of DNA databases is to solve crimes that would otherwise be unsolvable. While the material below is specific to the U.S. DNA database, similar concepts and metrics are used in other national DNA databases worldwide.

As mentioned in the previous section, a *hit* is a confirmed match between two or more DNA profiles discovered by the database search. Within CODIS, hits may occur at a local (LDIS), state (SDIS), or national (NDIS) level. Sometimes hits are distinguished as *"cold"* or *"warm"* depending on how much previous information is available when performing the search. A *cold hit* is one made without any prior investigative leads between the matching samples.

Both offender and forensic hits contribute to the bottom-line performance metric of a DNA database—the number of criminal *investigations aided*. The term "investigations aided" is defined as the number of criminal investigations to which CODIS has added value to the

TABLE 8.5 Hit Counting Statistics (Cumulative Totals by Year). Most Offender Hits (≈87%) Are Intrastate Rather than Interstate (National).

Year Ending Dec 31	Investigations Aided	Forensic Hits	Offender Hits	Within State Hits (≈87%)	National Offender Hits
2000	1573	507	731	705 (97%)	26
2001	3635	1031	2371	2204 (93%)	167
2002	6670	1832	5032	4394 (87%)	638
2003	11,220	3004	8269	7118 (86%)	1151
2004	20,788	5147	13,855	11,991 (87%)	1864
2005	31,485	7000	22,495	19,620 (87%)	2875
2006	45,364	9493	34,535	30,138 (87%)	4397
2007	62,725	11,890	50,244	43,688 (87%)	6556
2008	81,875	14,353	67,641	59,122 (87%)	8519
2009	105,918	17,935	89,798	78,581 (88%)	11,217
2010	129,514	21,790	112,742	97,190 (86%)	15,552

Source: CODIS brochure available at http://www.fbi.gov/about-us/lab/codis/codis_brochure and FBI Laboratory's CODIS Unit.

process (CODIS can only aid an investigation one time). In the first five years of operation (1998 to 2003), the CODIS system aided more than 11,000 investigations in the United States. Through the end of 2010, there have been more than 129,500 investigations aided using CODIS, allowing thousands of crimes to be linked and solved around the United States (Table 8.5).

Table 8.5 summarizes the growth of forensic and offender hits over the past few years. Note that almost 90% of the time, offender hits are occuring within states rather than between states, which emphasizes the value of DNA databases on a local level.

Because the number of hits is largely related to the size of the database, as the number of offender/arrestee profiles in NDIS continues to grow, there will be more investigations aided. The UK's NDNAD reportedly maintains close to a 40% chance of obtaining a match between a crime-scene profile and a "criminal justice" (arrestee or suspect) profile loaded into the database.

Number of DNA Profiles Compared with Different Searches

The number of DNA profiles being compared with different types of searches varies based on the number of DNA profiles contained in each index represented in Figure 8.2 and the types of searches performed (e.g., new forensic profile vs. entire offender index or new forensic profiles vs. entire forensic index).

Figure 8.3 illustrates the range of comparisons that may be made in forensic DNA analysis. The number of comparisons being made in each example is based on the number of crime scene samples in the forensic index (C) and the number of samples in the offender index (N).

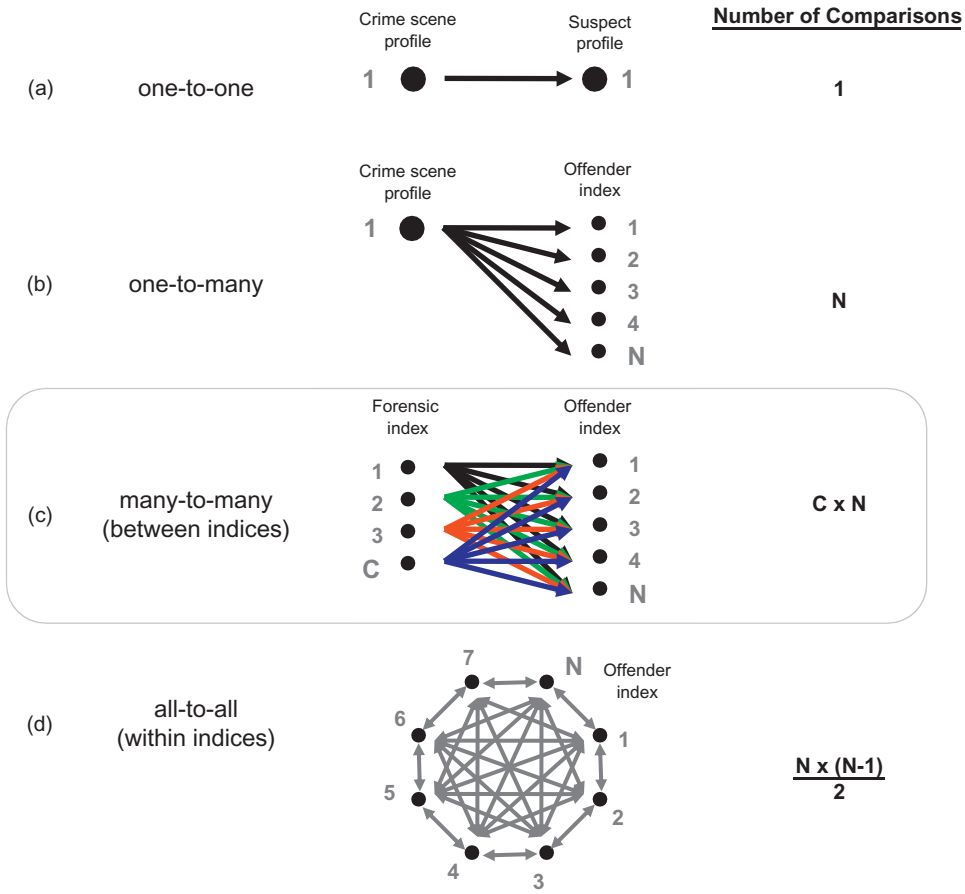


FIGURE 8.3 Illustration of various types and numbers of comparisons that may be performed in forensic DNA analysis with each dot representing a DNA profile: ranging from (a) a crime scene to suspect one-to-one comparison, (b) a "keyboard search" with one-to-many, (c) the typical database search between two indices containing C and N profiles, and (d) an all-to-all "Arizona search" within an index for quality control purposes.

Traditional forensic casework, in which a single-source profile is obtained, involves a one-to-one comparison of the question (Q) sample from the crime scene versus a single known suspect (K) sample (Figure 8.3a). Of course, if a mixture is obtained from the crime scene sample, then DNA profiles from multiple suspects or suspect(s) and victim may need to be evaluated.

If a single crime scene sample is compared to an offender index containing N individuals, then N comparisons are made (Figure 8.3b). This type of search is often called a "keyboard search." However, DNA databases contain many crime scene profiles that need to be compared to one another (requiring $C \times (C-1)/2$ unique comparisons) to search for "forensic hits," or crime scene profiles that need to be compared to offender profiles ($C \times N$ comparisons) to search for possible "offender hits" (Figure 8.3c). In either case, the number of

comparisons performed in the search is equal to the product of the number of samples in each index being compared (e.g., either C × C or C × N).

As mentioned above, an all-to-all search within an index (Figure 8.3d) can be useful for quality assurance purposes. Although these offender-to-offender searches can help maintain data quality and prevent unnecessary duplicates in the database, they can become expensive in terms of computer resources (software, hardware, and wall-clock time) due to the sheer volume of comparisons. With a million samples in a state DNA database, almost half a trillion (1 million × 1 million/2 = 500 billion) STR profile comparisons are required. This type of all-to-all search is commonly referred to as an "Arizona search" for reasons that will be explained later in the chapter.

As DNA databases grow in size, they become more valuable as an intelligence tool, but they also become more of a challenge to search rapidly.

When and How NDIS Searches Are Conducted

While state and local database uploads and searches are conducted as needed, the U.S. national database (NDIS) operates on a weekly schedule with searches performed once a week and data uploaded from each state (SDIS) on specific days for the remainder of the week. This schedule permits information to flow without clogging the system and gives time for candidate hits to be confirmed each week.

At the NDIS level, an AutoSearch is run at the beginning of each week. In actuality, the entire database is not searched over and over again each week. Rather, any DNA profiles received at NDIS in the past week are searched against the sample DNA profiles already present in the database (see Figure 8.2). Thus, new crime samples (≈1000 per week) are searched against the entire offender and arrestee indices. In addition, these new crime samples are searched against other specimens in the forensic index to look for potential case-to-case hits. New offender samples (≈25,000 per week) are likewise searched against the forensic index.

In a database of 9 million offender DNA profiles (N = 9,000,000) and 350 thousand crime samples (C = 350,000), there will be N × C comparisons made in searches between the offender and forensic indices (Figure 8.3c). Therefore, with 1000 new crime samples, nine billion comparisons are being made each week (N × C = 9,000,000 × 1000 = 9 billion). With the 25,000 new offender samples added to the database, another 8.8 billion comparisons are also being made (N × C = 25,000 × 350,000 = 8.8 billion).

Any candidate hits made during these searches are then communicated to the appropriate State or Local CODIS Administrator. By mid-week, most of the SDIS and LDIS labs have queued up hit confirmation samples. Verification of results can then be performed so that matching DNA profiles are connected to the offender's name so that local law enforcement in the appropriate jurisdiction can follow-up.

Search Stringency Used

To add to the complexity of a search, search algorithms have to take into account that DNA profiles in the forensic index, which often come from challenging evidentiary material,

may be mixtures rather than single-source profiles or partial profiles due to degraded DNA (see Chapter 10) or PCR inhibition (see Chapter 4). With unresolved mixtures, multiple possibilities need to be permitted in profile matching algorithms. In addition, various LDIS or SDIS laboratories may be submitting samples to NDIS from different STR kits that amplify the same STR locus with different primers. Therefore, allele dropout could result with one STR kit being used versus another one (see Chapter 7).

In order to not miss the true connection between two DNA samples from the same source, search strategies must have the capability to permit two DNA profiles to "match" without 100% allele agreement. Lower stringency search algorithms may be used to address this issue. This type of lower stringency search takes longer than a perfect match search because more possibilities have to be considered during each comparison.

There are three levels of search stringency possible in the CODIS software: high, moderate, and low stringency. With *high stringency*, all available alleles match (i.e., are the same) in the two DNA profiles. A *moderate stringency* search requires all available alleles to match, but the two profiles can contain a different number of alleles. For example, the forensic index profile may include alleles 9,10,12 (from a mixture) while the candidate offender match has only alleles 9,12 (from a single source). Likewise, a 9,12 will match a 9,- where the 12 allele has dropped out either due to primer binding site mutation or degradation. Moderate stringency is the standard search configuration.

A *low-stringency* match occurs when one or more alleles match at a given STR locus when the two profiles are compared. Thus, a low-stringency match would exist between profiles containing alleles 8,11 and 11,14 at a given STR locus because the "11" is common to the two profiles. Low stringency searching is used in missing person related searches where a parent's DNA profile is being used to search for a child or vice versa.

Mixture Samples and Obligate Alleles

Forensic casework often results in DNA mixtures. Some of these mixtures cannot be separated into individual components (i.e., separate DNA profiles from the original people contributing to the sample). However, there may be some alleles in a mixture profile that have higher peak heights or are definitely not from the victim and have a high likelihood of being from the perpetrator. These alleles are termed "obligate" because they are required to be present in any DNA profile possibility returned from a database search. When these types of casework STR profiles are entered into the forensic index, obligate allele(s) may be designated to help reduce the pool of candidate genotypes that can be associated with the target profile being searched.

Entering a complex mixture containing many alleles into the forensic index can lead to many candidate matches, almost all of which will be spurious (see ENFSI 2010, recommendation #14). In order to reduce the chance of false positives or "adventitious matches" on a DNA database search, filters have been established to prevent inappropriate profiles from being entered. As will be discussed in the next section, the originating laboratory (i.e., the LDIS or SDIS lab submitting the mixture profile) has to follow up on any candidate hits to their casework unknown profile. Thus, the submitting lab needs to avoid putting "garbage" into a searching system that would generate mostly "garbage" in return.

Followup to Database "Cold Hits"

When a potential match is identified by the CODIS software, the laboratories responsible for the matching profiles are notified. They then contact each other to validate or refute the match. After the match has been confirmed by qualified DNA analysts, which may involve retesting of the matching convicted offender DNA sample, laboratories may exchange additional information, such as names and phone numbers of criminal investigators and case details. A hit is only counted once the candidate match is confirmed and it is determined that the match provided additional information to the case.

If a match is obtained with the convicted offender index, the identity and location of the convicted offender is determined and an arrest warrant procured. In many cases, the matching offender may already be behind bars.

PopStats: A Tool for Estimating the Rarity of DNA Profiles

As noted previously in the chapter, there is a population database provided to CODIS users with allele frequency information intended to represent major population groups found in the United States. These databases are used to estimate statistical frequencies of DNA profiles using the program *PopStats*. The PopStats estimates include random match probability, combined probability of exclusion/inclusion, and likelihood ratios.

In the U.S., Caucasians, African Americans, Southeast Hispanics, and Southwest Hispanics are the typical groups reported when estimating the rarity of a particular DNA profile. The STR allele frequencies used in PopStats were published in the May 2001 issue of the *Journal of Forensic Sciences* by Bruce Budowle and FBI colleagues in an article entitled "CODIS STR loci data from 41 sample populations" (Budowle et al. 2001).

Missing Persons

DNA databases can also play an important role in helping to identify missing individuals and aiding mass disaster reconstruction following a plane crash or terrorist activity (see Chapter 9). In these cases, DNA samples are often obtained from biological relatives that can be searched against DNA of remains recovered from a missing individual or a disaster site. Many states within the United States and nations around the world are beginning to establish missing persons databases to enable matching of recovered remains to their family members.

To aid missing persons investigations, CODIS has several indices that can store DNA profiles from both recovered remains and family samples that serve as references. Much of the data from missing persons investigations is in the form of mitochondrial DNA (mtDNA) sequences since this information can be successfully recovered from highly degraded samples. As noted in Chapter 14, use of mtDNA also enables access to a larger number of reference samples from maternal relatives of a victim.

CODIS 6.0 and future versions of the next generation CODIS software enable work with autosomal STR data, Y-STR, and mtDNA data to aid missing persons investigations and work with mass fatality samples (Bradford et al. 2010). As of early 2011, Y-STR data is only being databased and searched in connection with missing persons cases.

NDIS Procedures Board

Operational procedures are set by the NDIS Procedures Board that meets several times each year. The NDIS Procedures Board is currently composed of six state and local representatives, a representative elected by the state CODIS administrators, the Chair of the Scientific Working Group on DNA Analysis Methods (SWGDAM), and five FBI Laboratory representatives, of which four have voting privileges. The state and local representatives are selected based on region of the country, laboratory size, and the number of searchable DNA records contributed to NDIS.

The U.S. system works on the principle of shared governance with ownership of data at the local or state levels. An NDIS Custodian, who is part of the CODIS Unit within the FBI Laboratory, manages the entire system with input from an NDIS Board, which creates NDIS Procedures. State and local laboratories that are connected through the CODIS software sign memoranda of understanding (MOUs) with the FBI in order to get access to the software and to participate in data sharing.

Status as a public criminal justice agency and laboratory accreditation are requirements for connectivity to NDIS. State CODIS administrators (those running the state DNA databases) come together multiple times a year to provide input on the operations of the database. In addition, SWGDAM (see Chapter 7) has a CODIS subcommittee to provide additional guidance. Quality Assurance Standards audits and NDIS assessments are performed regularly to ensure quality and consistency in DNA operations and compliance with federal law and procedures for participation in the National DNA Index.

DNA DATABASE LAWS IN THE UNITED STATES

DNA databases work because most criminals are repeat offenders (McEwen & Reilly 1994, Langan & Levin 2002). If their DNA profiles can be entered into the system early in their criminal careers, then they can be identified when future crimes are committed. Serial crimes can also be linked effectively with a computerized DNA database. Ultimately, the value of the DNA database is in its ability to apprehend criminals who are not direct suspects in a case and to prevent further victims from crimes committed by those individuals. Within the United States, both federal and state laws impact the use of DNA databases.

Federal Laws

As mentioned earlier in the chapter, the FBI Laboratory received Congressional authority to establish NDIS with the DNA Identification Act of 1994. NDIS became operational in October 1998 with nine participating states. Table 8.6 summarizes several federal laws since 1994 that have expanded on what may be stored and searched at NDIS as well as expansions of the federal DNA database program or qualifying offenders/arrestees. The federal laws have expanded the authority for collection of DNA samples first from certain federal felony offenders, then to all federal felony offenders, and finally to all federal felony arrestees and federal detainees. Some of these laws described in Table 8.6 have also authorized an expansion of the scope of DNA profiles that may be uploaded to and searched at NDIS, to persons indicted for a crime and then to all samples legally collected, which would include arrestee profiles for states statutorily authorized to collect from arrestees.

TABLE 8.6 A Brief Review of the History of Federal U.S. Laws on DNA Databases.

Legislation	What Was Authorized
DNA Identification Act of 1994	FBI receives authority to establish a National DNA Index System (NDIS); NDIS becomes operational in Oct 1998 with 9 states participating
DNA Analysis Backlog Elimination Act of 2000	Authorizes collection of DNA samples from federal convicted offenders
USA Patriot Act of 2001	Amends the DNA Analysis Backlog Elimination Act of 2000 to include terrorism or crimes of violence in the list of qualifying federal offenses
Justice for All Act of 2004	DNA profiles from indicted persons permitted at NDIS, one-time "keyboard" search authorized; accreditation and audit for labs required; expansion to all felonies for federal convicted offenders; requires notification of Congress if new core loci are desired
DNA Fingerprint Act of 2005	Arrestee and legally collected samples permitted at NDIS; elimination of one-time "keyboard" search; expansion to arrestee and detainee samples for federal offenders
Adam Walsh Child Protection and Safety Act of 2006	Amends the DNA Analysis Backlog Elimination Act of 2000 to authorize the Attorney General to collect DNA samples from individuals who are facing charges or are convicted, in addition to individuals who are arrested.

In addition to enabling arrestees' and all other legally collected DNA samples to be stored and searched at NDIS, the DNA Fingerprinting Act of 2005 expanded collection of DNA at the federal level to all convicted offenders and non-U.S. citizens detained by the U.S. government. The FBI Laboratory restructured their DNA units in 2009 in order to accommodate the anticipated dramatic growth in the number of samples requiring analysis. By introducing automation and hiring additional staff, the FBI's Federal DNA Databasing Unit reached a capacity of 60,000 samples per month at the end of 2010 (see Chapter 17).

Crimes for Inclusion in a State DNA Database

By June 1998, all 50 states in the United States had passed legislation requiring convicted offenders to provide samples for DNA databasing. However, each state has different requirements as to what types of offenses are considered for DNA sample collection. In many states these requirements are changing over time to include more and more criminal offenses.

The requirements for having to provide a DNA sample range from strictly sex offenses to all felons to all arrestees. The trend has been to enact laws that require a DNA sample submission for any felony and is now expanding in many cases to arrested individuals. Table 8.7 includes a summary list of the qualifying offenses for entry into a state's DNA database and the number of states within the U.S. that fall into each category as of 1999, 2004, 2008, and 2010.

Some state DNA database statutes specify exactly how the sample will be collected while others simply require any biological sample containing DNA. A 1999 survey of state laws found that California required two specimens of blood, a saliva sample, and right thumb

TABLE 8.7 Summary of U.S. State DNA Database Laws and Qualifying Offenses for DNA Collection as of 1999, 2004, 2008, and 2010. For Up-to-Date Information, See http://www.dnaresource.com and http://www.ncsl.org.

	Number of States				Hit Rate Estimates*
Offenses	**1999**	**2004**	**2008**	**2010**	
Sex crimes	50	50	50	50	*5%*
All violent crimes	36	48	50	50	*10%*
Burglary	14	47	50	50	*20%*
All felons	5	37	47	49	*45%*
Juveniles	24	32	32	32	–
Arrestees	1	4	14	25	*60%*
Familial searching being actively used	–	–	2	2	–

**Hit rate estimates are from http://www.dnaresource.com and reflect the benefits of expanding the types of qualifying offenses and the DNA database size.*

and full palm print impressions to verify the identity of the submitting convicted offender, while South Carolina only asked for "a suitable sample" from which DNA may be obtained (Herkenham 1999).

The ability of state and local forensic DNA laboratories to improve their capabilities for DNA analysis, especially with the STR technology described in this book, has been greatly aided by federal funding. The DNA Identification Act of 1994 provided approximately $40 million in federal matching grants to aid states in DNA analysis activities. Additional Congressional allocations over the years have brought significant federal funding to state and local laboratories since the establishment of the U.S. national DNA database.

The Debbie Smith Act, part of the President's DNA Initiative, has brought hundreds of millions of dollars of funding to U.S. forensic DNA laboratories since 2004. This funding has been a great benefit to forensic DNA laboratories, which are often understaffed and underfunded. Progress on legislation regarding the use of DNA is available through the website http://www.dnaresource.com.

ISSUES AND CONCERNS WITH DNA DATABASES

There are a number of important issues for DNA databases. These issues include privacy and security of the information contained in the database, the ability to perform rapid searches and effective matching among large numbers of entries, and ensuring that high-quality data are submitted for both offender and forensic samples. In addition to maintaining the quality of the input data, database administrators must handle changes in technology. At the end of the chapter, further concerns commonly raised by DNA database critics will be addressed.

Privacy Concerns

One of the major challenges for maintaining a DNA database is the issue of privacy and security of the information stored since the database contains DNA profiles that correspond with a databank of biological samples. DNA samples contain genetic information that could be used against an individual or his or her family if not handled properly.

A conference held in Boston, Massachusetts on May 11–12, 2006 discussed many of the privacy and civil liberty concerns regarding DNA testing and DNA databases. The *Journal of Law, Medicine, & Ethics* published the proceedings of this "DNA Fingerprinting and Civil Liberties" conference in its summer 2006 issue. The titles and authors of the articles are included in Table 8.8.

The issue of privacy can be approached in multiple ways. First, the DNA markers, such as the 13 CODIS core STR loci, are in non-coding regions of the human genome and are not known to have any association with a genetic disease or any other genetic predisposition (Butler 2006, Kaye 2007). Thus, the information in the database is only useful for human identity testing (see D.N.A. Box 8.1).

Second, no names of individuals or other characterizing data are stored with the DNA profiles at the national level. The national level of CODIS only references the sources of the DNA profiles, such as Orange County Sheriff's Office or Palm Beach County Crime Laboratory. Specific case data are secured and controlled by the law enforcement agencies that submit the data. Thus, only the crime laboratory that submitted the DNA profile has the capability to link the DNA results with a known individual.

Third, data are encrypted and shared through a secure network only accessible to state and local CODIS administrators.

Fourth, federal and state penalties for improper use of criminal DNA samples include fines and possible imprisonment. Access to CODIS is solely for law enforcement purposes. There are strict penalties for anyone who uses the information or samples for any purpose other than for law enforcement. The penalties include a $250,000 fine for unauthorized disclosure of information on any sample (Herkenham 2006).

Handling Technology Changes and Legacy Data

Both computer and DNA technologies are evolving at a rapid rate. DNA databases have to be flexible enough to handle these changes. In particular, legacy data must be maintained, or the value of the database will be diminished. If different genetic markers are universally used at some time in the future, then previously collected data would be inaccessible unless there is an overlap or compatibility between the different marker systems (Gill et al. 2004). Typing potentially millions of previously tested samples with new genetic markers would be an expensive and time-consuming proposition. There are several possible ways forward to maintain core loci while adding auxillary information to established DNA databases.

As has been noted many times throughout this book, DNA testing is always a matter of comparing a question (Q) sample against a known (K) sample. This Q-K comparison requires that both samples be tested with the same genetic markers. Thus, if a new technology or set of genetic markers become available that enables improved recovery of information from crime scene samples, it would require retroactive analysis of all previously

TABLE 8.8 Authors and Titles of Articles Appearing in the Summer 2006 Issue of the *Journal of Law, Medicine, & Ethics*.

Author(s)	Page numbers	Article Title
Benjamin W. Moulton	147–148	DNA Fingerprinting and Civil Liberties
Alice A. Noble	149–152	DNA Fingerprinting and Civil Liberties
Mark A. Rothstein, Meghan K. Talbott	153–164	The Expanding Use of DNA in Law Enforcement: What Role for Privacy?
Tracey Maclin	165–187	Is Obtaining an Arrestee's DNA a Valid Special Needs Search Under the Fourth Amendment? What Should (and Will) the Supreme Court Do?
David H. Kaye	188–198	Who Needs Special Needs? On the Constitutionality of Collecting DNA and Other Biometric Data from Arrestees
Tania Simoncelli, Barry Steinhardt	199–213	California's Proposition 69: A Dangerous Precedent for Criminal DNA Databases
Amitai Etzioni	214–221	A Communitarian Approach: A Viewpoint on the Study of the Legal, Ethical and Policy Considerations Raised by DNA Tests and Databases
Frederick R. Bieber	222–233	Turning Base Hits into Earned Runs: Improving the Effectiveness of Forensic DNA Data Bank Programs
Robin Williams, Paul Johnson	234–247	Inclusiveness, Effectiveness and Intrusiveness: Issues in the Developing Uses of DNA Profiling in Support of Criminal Investigations
Henry T. Greely, Daniel P. Riordan, Nanibaa A. Garrison, Joanna L. Mountain	248–262	Family Ties: The Use of DNA Offender Databases to Catch Offenders' Kin
Erica Haimes	263–276	Social and Ethical Issues in the Use of Familial Searching in Forensic Investigations: Insights from Family and Kinship Studies
Pilar N. Ossorio	277–292	About Face: Forensic Genetic Testing for Race and Visible Traits
Troy Duster	293–300	Explaining Differential Trust of DNA Forensic Technology: Grounded Assessment or Inexplicable Paranoia?
Mervyn L. Tano	301–309	Interrelationships among Native Peoples, Genetic Research, and the Landscape: Need for Further Research into Ethical, Legal, and Social Issues
Paul C. Giannelli	310–319	Forensic Science
Margaret A. Berger	320–327	The Impact of DNA Exonerations on the Criminal Justice System
Sheila Jasanoff	328–341	Just Evidence: The Limits of Science in the Legal Process

(*Continued*)

TABLE 8.8 Authors and Titles of Articles Appearing in the Summer 2006 Issue of the *Journal of Law, Medicine, & Ethics*. (*Continued*)

Author(s)	Page numbers	Article Title
Gregory Carey, Irving I. Gottesman	342–351	Genes and Antisocial Behavior: Perceived versus Real Threats to Jurisprudence
Bartha Maria Knoppers, Madelaine Saginur, Howard Cash	352–365	Ethical Issues in Secondary Uses of Human Biological Materials from Mass Disasters
David Lazer, Viktor Mayer-Schönberger	366–374	Statutory Frameworks for Regulating Information Flows: Drawing Lessons for the DNA Data Banks from other Government Data Systems
R. E. Gaensslen	375–379	Should Biological Evidence or DNA be Retained by Forensic Science Laboratories After Profiling? No, Except Under Narrow Legislatively-Stipulated Conditions
M. Dawn Herkenham	380–384	Retention of Offender DNA Samples Necessary to Ensure and Monitor Quality of Forensic DNA Efforts: Appropriate Safeguards Exist to Protect the DNA Samples from Misuse
Michael E. Smith	385–389	Let's Make the DNA Identification Database as Inclusive as Possible
Tania Simoncelli	390–397	Dangerous Excursions: The Case Against Expanding Forensic DNA Databases to Innocent Persons
Lori Andrews	398–407	Who Owns Your Body? A Patient's Perspective on *Washington University v. Catalona*
Sharon F. Terry, Patrick F. Terry	408–414	A Consumer Perspective on Forensic DNA Banking

collected and typed forensic and offender samples (Gill et al. 2004, Gill et al. 2006a). Information not compatible with previously typed samples would render those existing DNA databases obsolete. Thus, what will likely happen in the future is to extend sample testing to additional loci but keep all or a portion of previously typed markers (Figure 8.4).

In many cases, the extension to additonal STR loci illustrated in Figure 8.4b is already happening with the analysis of STR kits like PowerPlex 16 or Identifiler, which include the 13 CODIS core STR loci plus two additional loci (e.g., Penta D and Penta E with PowerPlex 16 or D2S1338 and D19S433 with Identifiler). The ability to create large multiplex PCR assays is critical to being able to extend the reach of information collected on new loci while retaining original loci.

The scenario illustrated in Figure 8.4c is also possible because some of the original core STR loci are not very informative. For example, CSF1PO, TPOX, and TH01 are not as polymorphic as most of the other core STRs used in the U.S. They were originally selected for historical reasons as they were some of the first STRs analyzed (see Chapter 5). Now that many more STR loci have been characterized with higher powers of discrimination, replacing some of these early STRs with other loci is an option worth considering. Of course,

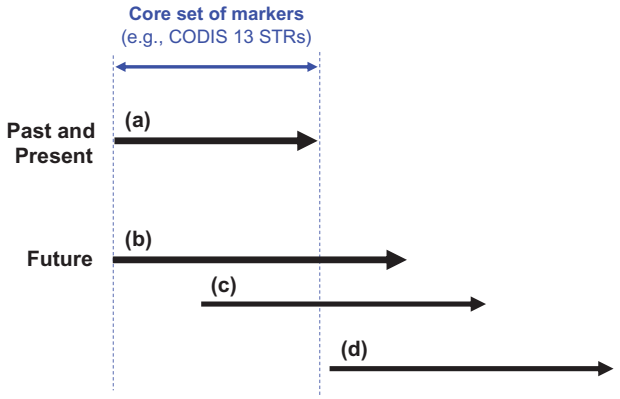


FIGURE 8.4 Possible scenarios for extending sets of genetic markers to be used in national DNA databases. (a) A core set of markers have been established for past and present DNA profiles now numbering in some cases in the millions of profiles (e.g., in the U.S., 13 CODIS core STR loci). Three future scenarios exist: (b) keep all of the current core loci and add some additional supplemental loci (e.g., Identifiler or PowerPlex 16 add two additional STRs), (c) only retain some of the current core loci and add more additional supplemental loci, or (d) abandon the previous genetic markers and have no overlap with current core loci.

the possibility of removing loci from future multiplex STR assays will have to be weighed against the value of retaining a connection to previous data from these markers.

What is unlikely, at least for applications involving criminal DNA databases in the foreseeable future, is the abandonment of previously typed genetic markers in favor of a whole new set (Figure 8.4d). However, new loci or assays could be used in other situations, such as parentage testing or disaster victim identification. For these applications, comparisons are made within a closed population data set. With a criminal DNA database, comparisons must be made across a much broader past, present, and future data set.

The expense of replacing previously typed samples from both offenders and crime scene evidence with new markers will likely mean that most countries will continue to work with legacy sets of genetic markers and will follow either of the Figure 8.4b or 8.4c scenarios. New STR typing kits will most likely become available to aid forensic DNA testing of the future with an overlap of old and new STR loci.

Working Unknown Suspect Cases

Crime laboratories must work cases that have no suspect in order to take full advantage of DNA databases. Convicted offender samples can be typed in large batches because large numbers come into the laboratory together and they are in the same format, such as liquid blood or buccal swabs.

Casework samples, on the other hand, present a different kind of challenge. Each case requires significant up-front work including evidence handling, locating a biological stain within the submitted evidence, and extraction of DNA from different types of substrates. Often sample mixtures must be dealt with and interpreted. Multiple pieces of evidence may also be involved in a case. In addition, significant work is required after analysis of

the samples. For example, laboratory reports must be written and court testimony may be required once a suspect has been identified.

In spite of the time and effort required to obtain results on crime scene samples, it is these cases that make DNA databases effective. Law enforcement agencies must be encouraged to collect and submit evidence to the nation's crime laboratories especially if the statute of limitations is about to expire on a case (see information on "John Doe" warrants below). There have been reports of thousands of rape kits that are not submitted to crime laboratories but are sitting in police evidence rooms (Lovirich et al. 2004).

Backlogs

A National Institute of Justice-funded review of police evidence rooms published in 2004 found that over 540,000 cases with potential biological evidence were either still in the possession of local law enforcement or backlogged at forensic laboratories (Lovirich et al. 2004). Funding from the President's DNA Initiative (see http://www.dna.gov) has helped reduce these backlogs but new ones have arisen as the efforts to use DNA to solve crimes have expanded.

In a more recent report, the U.S. Department of Justice Office of Inspector General found that in March 2010, the FBI Laboratory had a backlog of 3211 forensic DNA cases (OIG 2010). Without taking on any new cases or adding new staff, the report estimated that it would take the FBI about two years to eliminate its existing forensic DNA case backlog. These backlogs are growing in many laboratories—by 757 cases for the FBI over an 18 month time period in 2009–2010 (OIG 2010). The existence of backlogged cases slows reporting of results to investigators—some case contributors have to wait more than 600 days for results from the FBI Laboratory (OIG 2010). Clearly, improving sample throughput is important if DNA cases are going to be handled in a timely manner. Capacity building through hiring additional personnel and making process improvements are crucial to reducing delays in completing cases.

NIJ published a special report in June 2010 entitled "Making sense of DNA backlogs—myths vs. reality" that addresses why DNA backlogs still exist despite the federal government investing hundreds of millions of dollars on this issue (Nelson 2010). This report concludes that "crime laboratories have increased their capacity to work cases significantly, but they are not able to eliminate their backlogs because the demand continues to exceed the increases made in capacity" (Nelson 2010).

The California Department of Justice Jan Bashinski DNA Laboratory publishes monthly statistics on its sample backlog and the progress made with adding new samples to the state DNA database (http://ag.ca.gov/bfs/pdf/Monthly.pdf). Data points from three months are included in Table 8.9 to illustrate the trends observed when trying to reduce backlogs.

Through a great deal of effort on the part of this DNA laboratory, the November 2006 starting backlog of 221,052 samples was reduced to 39,651 by November 2010. However, the ending backlog for November 2010 was 2028 samples larger (41,679 compared to 39,651) than the starting backlog for the same month because about 10% more samples were coming into the laboratory than could be processed that month. It is noteworthy that as the California SDIS database has grown in size so has the number of hits per month (e.g., 201 in November 2006 to 343 in November 2010).

TABLE 8.9 Three Time Points in the DNA Backlog Results from California's Jan Bashinski DNA Laboratory. For the Most Recent Data, See http://ag.ca.gov/bfs/pdf/Monthly.pdf.

Month	November 2006	July 2009	November 2010
Starting Backlog	221,052	61,611	39,651
Ending Backlog	197,227	60,815	41,679
Total Offender Profiles in SDIS	662,542	1,294,314	1,660,025
Total Forensic Unknowns in SDIS	14,813	26,887	35,800
Hits (that month)	201	317	343
Total Hits (cumulative)	3346	9701	14,925

Automation of sample processes—especially data review—helps speed analysis and increase sample throughput. Several expert systems have been approved to automatically review DNA profiles that are then uploaded to SDIS and NDIS (Table 8.4). These expert systems include the FSS-i^3 software available from Promega Corporation (Madison, WI), GeneMapper*ID* and GeneMapper*ID-X* available from Applied Biosystems (Foster City, CA), and TrueAllele from Cybergenetics (Pittsburgh, PA). At the National CODIS Conference in November 2010, the FBI's NDIS Custodian stated that, at that time, over one million of the almost nine million profiles in the national database had been reviewed with expert systems. Improved efficiency in data analysis is an important reason that the U.S. national database is growing more quickly in recent years.

Sample Collection from Convicted Offenders

One of the aspects of DNA databanks that is often overlooked is the sample collection process. Law enforcement personnel have to extract blood or obtain a saliva sample from incarcerated felons who are not always cooperative. In some cases, extraordinary efforts including force may be required to persuade felons to submit to sample collection (Spalding 1995). Collecting the actual samples can be a challenge considering the fact that the convicted offender likely knows his blood or saliva could be used to catch him committing another crime in the future or match him to a previous unsolved crime that he committed.

Unfortunately, there can be poor tracking of sample collection or incomplete communication between the prisons, probation officers, and DNA laboratory that result in failure to collect all of the appropriate DNA samples. In some situations, efforts are mounted to go back and get a DNA sample collected from *everyone* who qualifies. For example, the state of Wisconsin discovered that over a 16-year timeframe from August 1993 to September 2009, some 17,698 samples had not been collected from a total of 130,368 offenders who were required to provide a sample (DNA Task Force Report 2010). They are making every effort to recover the missing samples to make the database complete as required by the Wisconsin legal statutes and expected by the general public.

Duplicate Samples or Twins

Good communication is crucial between those agencies that collect offender and arrestee samples and the DNA laboratory that processes the samples. Due to the nature of recidivism among criminals and the fact that they are often given short stays in prison, a reference sample from the same individual may be submitted to a DNA database multiple times—particularly if multiple aliases are used when samples are collected at different points in time. Duplicates do occur in the offender index. Sometimes they are removed when further investigation reveals that the DNA profiles are from the same individual. For this reason, fingerprints are often used to verify that the DNA profile is from the same individual and not an identical twin. (Monozygotic or identical twins have the same DNA profile but different fingerprints.) Without an accompanying biometric like fingerprints to pre-screen the DNA samples, duplicate submissions may be inadvertently analyzed by the DNA database laboratory, thereby wasting scarce resources and time.

Matches to one member of a set of identical twins can cause problems for the prosecution, as in a 2009 case in Germany where authorities had to let two thieves go free after a DNA database hit matched both of them (Himmelreich 2009a). The court ruling stated in part: "From the evidence we have, we can deduce that at least one of the brothers took part in the crime, but it has not been possible to determine which one."

While there is no record kept on a national level for numbers of identical twins within the U.S. database (because no names or personal identifying information is kept on any of the NDIS samples), there have been a few reports from other national DNA databases. After its first four years of operation, the Swiss database contained 35 monozygotic twin pairs among a total of 61954 DNA profiles (Voegeli et al. 2006). As of March 31, 2007, the UK NDNAD reported 2914 sets of identical twins and 4 sets of identical triplets on their database of 3,874,500 individuals (NPIA 2007).

Sample Retention

Critics of DNA databases and databanks are concerned that DNA samples from offenders are retained after a DNA profile of STR markers has been generated from the sample (Gaensslen 2006, van Camp & Dierickx 2008). As has already been noted, the justification of retaining these samples is two-fold (Herkenham 2006). First and foremost, because samples are retested following an offender hit, it is necessary to have the original sample to confirm its STR profile. This hit confirmation is an important quality control measure and helps prevent any kind of potential error during laboratory processing that could cause a lead to be followed and a warrant to be issued for the wrong person.

As noted by the NRC II report, "the ultimate safeguard against error due to sample mixup is to provide an opportunity for retesting" (NRC 1996, p. 81). While we typically think of this statement in terms of evidentiary items from casework, it applies equally well to DNA database samples.

Another reason to retain sample specimens is to enable analyses using technology advancements in the future. If new genetic markers or assays are developed in the future (see Figure 8.4) to enable better recovery of information from forensic samples, then previously examined forensic evidence and offender samples would need to be retested with

the new genetic markers/technology in order to permit a comparison between offender and forensic results.

Challenges with Sample/DNA Profile Expungement

If an arrested and/or convicted individual is later cleared of their charges, they may want their DNA profile removed from the database and their DNA sample destroyed or returned from the databank. However, from the laboratory's standpoint, the removal of electronic records and physical samples may not be trivial.

Forensic DNA laboratories generally do not have a link to any criminal justice or court databases. In most jurisdictions, procedures are not in place to track whether or not someone was acquitted during a trial or if charges were dropped during the judicial process. Automatic expungement is not really an option with the current system. Thus, the individual is typically asked to write the law enforcement agency, to request the removal of his/her DNA profile and the destruction of the DNA sample, and to provide appropriate documentation to support his/her request. For example, the FBI's CODIS expungement policy is available at http://www.fbi.gov/about-us/lab/codis/codis_expungement.

Measuring DNA Database Performance

Measuring success is important in any endeavor. A commonly used metric with DNA database searches is the "hit" that results from a match between an offender and an unknown DNA profile from a forensic case. As mentioned earlier, there are "cold" hits and "warm" hits that depend on whether or not prior information is available about a potential contributor. The growth of DNA databases has been driven by some spectacular success stories and data on the types of crime solved.

DNA databases are effective for identifying serial offenders. For example, in 2004 Fletcher Worrell (age 58 at the time) tried to purchase a shotgun in Atlanta, Georgia. His background check revealed two arrest warrants in New York City from 1978. After his DNA sample was collected and analyzed and his profile uploaded, it resulted in NDIS hits that matched 21 unsolved cases in Maryland (from the so-called "Silver Spring Rapist") and 8 unsolved cases from other states. Thus, the DNA database enabled 31 cases to be solved spanning more than 25 years! Some other success stories are available at http://www.dna.gov/case_studies/preventable/.

In 2006, the Swiss DNA database reported on their performance during the first four years of operation (Voegeli et al. 2006). A total of 61,954 DNA profiles had been collected using the SGM Plus kit, which contains 10 STR loci and amelogenin. Of these profiles, 53,400 were from suspects (89% male and 11% female) and 8,554 were from forensic stains (90.7% single-source, 9.3% mixtures). The qualifying offense for the suspects that resulted in the first 6,830 hits included 85% burglary, 4.2% homicide, 4% robbery, 2.4% sexual offenses, and 4.2% from other crimes such as arson, fraud, and illicit drug use. While most of the hits linked the offender to a single crime, in one case 62 different crime stains were linked to a single perpetrator through DNA testing.

In their report, the Swiss DNA laboratory noted that over this four-year period, 36 stains were matched to police or laboratory staff and appropriately detected as contamination that

occurred during sample collection or analysis (Voegeli et al. 2006). They effectively avoided uploading these DNA profiles to the database because they searched their elimination database (see Chapter 4) regularly against all evidentiary results.

An analysis of the number of hits obtained over a particular time period can be instructive as to the types of offenses where DNA testing is making a difference. New York State found that its first 1,000 DNA database hits (see http://www.criminaljustice.state.ny.us/forensic/typesofcrimesfirst1000hits.htm) resulted in solving 730 sex offenses, 135 burglaries, 65 murders/attempted murders, 28 robberies, 17 assaults, and 25 "miscellaneous" crimes. Virginia estimates that 80% of their hits would have been missed if DNA collection was only limited to violent offenders. Their analysis also found that 40% of violent crimes are perpetrated by individuals who have previous property crime convictions (see http://www.dfs.virginia.gov/statistics/index.cfm).

In an effort to help normalize database output or efficiency and thereby gain a better idea of how input of information into a DNA databases impacts its output, Simon Walsh from the Australian Federal Police and colleagues from New Zealand have developed some models for measuring database output and efficiency (Walsh et al. 2010). The goal of their performance metrics is to see how well DNA databases perform with the idea that this information could be used to influence policy regarding the types of samples that should be collected and analyzed (D.N.A. Box 8.2).

Followup to Database Matches

On November 20, 2006 *USA Today* ran a story discussing the lack of follow-up to many DNA cold hits (Willing 2006). Even though a DNA hit has been observed and reported, police and prosecutors may not pursue the lead provided by a CODIS hit. Thus, the cold case stays cold in spite of the efforts of DNA scientists.

There are at least four general reasons that a DNA database hit does not result in a conviction of the individual matched to the evidentiary sample (Bieber 2006): (1) variable follow-up after the hit is reported to the police, (2) witness/suspect issues, (3) time, and (4) trial/evidentiary issues.

First, detectives and prosecutors may be overloaded with current cases and may not make an old/cold case a priority. Also, there may be a breakdown in reporting the hit to the appropriate law enforcement personnel.

Second, victims, police, and other witnesses involved in a cold case may have died, may be unable to be located, or may be unwilling to testify in an old case. Suspects may be dead, unable to be located, or already incarcerated. If a suspect is already in prison, then the investigators and prosecutors may not feel a need to attempt to convict the suspect for the crime connected with the DNA hit.

Third, statutes of limitations may have expired or victims may have put the crime behind them and moved on with their lives and are thus unwilling to open old wounds and revisit a difficult event from their past.

Finally, the DNA evidence may not necessarily be considered probative or otherwise admitted in trial. In this case, the jury may acquit if the prosecution is not able to make a strong enough case against the suspect without using the DNA evidence.

D.N.A. BOX 8.2

MODELING DNA DATABASE PERFORMANCE

With the rapid growth of DNA databases due to expanded legislation, the answers to several questions might be desired, such as (1) Are databases becoming more effective as they grow in size?, or (2) What is the impact of adding particular sets of individuals (such as arrestees)?

Simon Walsh of the Australian Federal Police, along with New Zealand statistician colleagues James Curran and John Buckleton, have developed a measure of DNA database performance in an effort to equate hit efficiency with the types of samples going into the database.

As was illustrated in Figure 8.2 and Figure 8.3c, hits are made as N offenders are compared against C crime stain profiles. The Walsh et al. (2010) model is that

$$H = \frac{\alpha N}{M} \times \omega C$$

where H is the number of DNA database hits (matches), N is the number of DNA profiles in the offender index, M is the active criminal population, C is the number of DNA profiles from crime scenes in the forensic index, α (alpha) is a quality factor related to person sampling, and ω (omega) is a quality factor related to crime scene sampling.

According to this model, the best return on a database (i.e., generating the highest number of possible hits) is (1) having an optimal ratio of N offender profiles to C forensic profiles, (2) having a low active criminal population, (3) having a high α value which relates to a sample collection strategy that effectively gathers the active criminals, and (4) having a high ω value which relates to a forensic sampling strategy that obtains the best samples associated with the crime and avoids irrelevant profiles.

Source:

Walsh, S.J., et al. (2010). Modeling forensic DNA database performance. Journal of Forensic Sciences, *55, 1174–1183.*

In one of the most thorough studies to-date, the San Francisco Police Department (SFPD) Criminalistics Laboratory reviewed case resolutions of 198 DNA database hits observed in their laboratory in order to evaluate the impact of their work on public safety (Gabriel et al. 2010). Their 198 LDIS hits came from 164 offender hits and 34 case-to-case hits with offenses involving 24 homicides, 110 sex crimes, 42 burglaries, and 22 other crimes. The SFPD scientists examined 12 felons in greater detail; collectively, these criminals had committed 199 offenses that led to arrests. Unfortunately, the criminal justice system was unable to successfully interrupt their criminal behavior patterns over time. For example, 46% of the sexual offenses in the cold hit cases reviewed had occurred while these 12 prolific felons were serving probation for prior convictions.

In some cases, the DNA hits helped strengthen the prosecution's case and increase the jail sentence. The SFPD scientists conclude that there need to be better tools for local jurisdictions to track cold hits and criminal case outcomes so that connecting DNA information does not slip through the cracks. They observed that 48% of cold cases were dismissed, typically because of victims declining to pursue prosecution. However, it was noted that not

a single case that went to court resulted in aquittal by jury trial. In addition, 90% of property crime cases resulted in a CODIS-eligible profile with a 72% corresponding cold hit rate (Gabriel et al. 2010).

Potential of Adventitious Matches with Fewer Loci or Database Growth

An "adventitious" match is one that happens by chance instead of design. In the case of DNA testing, not having enough distinguishing characteristics (e.g., due to a partial profile) could lead to adventitious matches. As DNA databases grow in size and more comparisons are made, the potential exists for adventitious matches to occur unless additional loci are added to provide a finer resolution or increased ability to distinguish DNA profiles from one another. Examining additional loci in a sample will increase the random match probability. The ENFSI document on DNA database management relates the size of a DNA database to the potential for an adventitious match (D.N.A. Box 8.3).

The only published account of an adventitious (false positive) match from a DNA database came in 1999, when the UK database, then consisting of 660,000 profiles with only 6 STR loci (SGM assay), led to a "hit" between two individuals whose 6-locus random match probability was 1 in 37 million (Willing 2000). Further testing with four additional STRs (SGM Plus loci) showed that the samples were from different individuals. Around the same time, the UK expanded the number of core loci from 6 to 10 with the adoption of the SGM Plus kit to try to prevent another adventitious match.

The growth of DNA databases necessitates the inclusion of additional loci to avoid this problem. For this reason, in 2009, the European community expanded their core set of loci from 7 to 12 (see Chapter 5). As of late 2010, the U.S. is also considering expanding the number of required loci beyond the current 13 CODIS STRs to reduce the chance of future adventitious hits with database growth, to improve success with missing person investigations, and to make international data comparisons more feasible as needed.

Partial Matches and the "Arizona Search"

As mentioned previously, DNA profiles from relatives may be present in DNA databases. Therefore, adventitious hits could be observed from a portion of a DNA profile in the forensic index matching a relative of the true source of the sample found in the offender index. At the International Symposium on Human Identification (aka the Promega meeting) in 2000, a member of the Italian National Police noted an 8 STR locus match between two brothers in what was essentially an early familial search (Biondo 2000).

However, it was a poster presented the following year in October 2001, by a scientist from the Arizona Department of Public Safety that gained much greater attention. Kathryn Troyer's poster was entitled "A nine STR locus match between two apparently unrelated individuals using AmpFlSTR Profiler Plus and COfiler" (Troyer et al. 2001). Their abstract reads:

> *The Arizona Department of Public Safety Crime Laboratory is reporting a nine STR locus match between two apparently unrelated individuals at vWA, D21S11, D5S818, D13S317, D7S820, D16S539, TH01, TPOX, and CSF1PO. The samples were analyzed by Myriad Genetic Laboratories, using Applied Biosystem's AmpFlSTR Profiler Plus and COfiler kits. The nine STR locus match was discovered using a duplicate search by Myriad Genetic Laboratories. Further*

D.N.A. BOX 8.3

ESTIMATES OF ADVENTITIOUS MATCHES BASED ON DNA DATABASE SIZE AND MATCH PROBABILITY OF DNA PROFILES

As DNA databases add profiles and grow in size, there becomes a greater probability that two profiles will match by chance rather than because they originate from a common source. This false positive match is called an *adventitious match*. The best way to avoid an adventitious match when the size of the DNA database grows is to increase the overall "1 in X" average random match probability (RMP) for the profiles found within the database. Increasing this value is accomplished by adding more loci to the DNA profile. Alternatively, loci can be removed in simulation experiments to examine the impact of adventitious matches with partial profiles (Hicks et al. 2010).

For the past several years, the European Network of Forensic Science Institutes (ENFSI) DNA Working Group has annually prepared a document on DNA database management. Below is one of the tables from the 2010 document illustrating the expected number of adventitious matches when searching a DNA database of given size (horizontal axis) with a DNA profile having a certain RMP (vertical axis). Ideally, for the DNA profiles contained in the database, the "1 in X" RMP should be high enough that adventitious matches are avoided (i.e., much less than 1).

Sources:

ENFSI DNA Working Group (2010). DNA-Database Management: Review and Recommendations. Available at http://www.enfsi.eu/get_doc.php?uid=345.

Hicks, T., et al. (2010). Use of DNA profiles for investigation using a simulated national DNA database: Part I. Partial SGM Plus profiles. Forensic Science International: Genetics, *4, 232–238.*

	Size of the DNA-database				
		10,000	**100,000**	**1,000,000**	**10,000,000**
Random *P* Match Probability (1:X)	10,000	1	10	100	1,000
	100,000	0.1	1	10	100
	1,000,000	0.01	0.1	1	10
	10,000,000	0.001	0.01	0.1	1
	100,000,000	0.0001	0.001	0.01	0.1
	1,000,000,000	0.00001	0.0001	0.001	0.01
	10,000,000,000	0.000001	0.00001	0.0001	0.001

investigation by the Arizona Department of Public Safety revealed that the two offenders varied by age and race. In addition, it was noted that the two offenders shared one allele at three of the remaining four loci, D3S1358, D8S1179, and D18S51. The match is in the process of being confirmed with Applied Biosystem's AmpFlSTR Identifiler kit.

After comparing all offender profiles to all other profiles in the database, a partial match was identified between two different offender profiles in which 9 out of 13 loci had the same genotype. The type of search where *all samples* are compared to *all other samples* in a DNA database is commonly referred to now as an "Arizona search" (see Figure 8.3d). Often this type of search also includes looking for how many samples may match at 9, 10, 11, or 12 loci instead of the full 13 CODIS STRs.

In a 2004 *Journal of Forensic Sciences* article, Professor Bruce Weir noted that partial matches occur when examining profile comparisons in large data sets (Weir 2004). Not being aware of the Arizona search example, he states in his paper: "As offender databases grow…high degrees of matching are to be expected. It is very likely that there are already 9-locus matches within combined U.S. offender databases" (Weir 2004).

This partial match issues has been presented as being similar to the so-called "birthday problem" (Budowle et al. 2006, Budowle et al. 2009), which is described in D.N.A. Box 8.4. Keep in mind that the chance of a *specific* DNA profile matching to another DNA profile is a very different situation than the chance of *any* two DNA profiles matching.

Database Match Probabilities for Cold Hit Statistics

There is sometimes confusion between the *random match probability* (RMP) of a DNA profile and what is termed the *database match probability* (DMP). The 1996 National Research Council report (NRC II) recommendation 5.1 states that to determine the database match probability the random match probability should be multiplied by N, the number of persons in the database. The FBI's DNA Advisory Board endorsed the NRC II calculation in their February 2000 recommendations on statistical approaches.

While the DMP is easy to calculate once the RMP is determined, it is not usually put in case reports. Mentioning a DMP calculation during expert witness testimony may cause issues in court since having a DNA sample in a database would show prior conviction. This information could then possibly bias a jury against a defendant or lead to a mistrial.

Most of the confusion arises from the fact that these RMP and DMP values are answers to different questions (Chakraborty & Ge 2009). The RMP answers the question "What is the rarity of a specific DNA profile given the alleles observed?" The DMP answers the question "How often would a DNA profile match the relevant forensic sample in a database of size N?" The DMP relates to the types of searches illustrated in Figure 8.3c. The differences between RMP and DMP have been debated in several court cases—and several helpful articles have been written trying to make this topic clearer (Storvik & Egeland 2007, Kaye 2008, Chakraborty & Ge 2009).

Desire for Academic Research on NDIS Data

In December 2009, the journal *Science* published a letter to the editor from Dan Krane of Wright State University (Dayton, Ohio) and 40 (mostly) academic colleagues requesting that

D.N.A. BOX 8.4

THE "BIRTHDAY PROBLEM" AND THE IMPORTANCE OF HOW YOU FRAME THE QUESTION YOU WANT ANSWERED

The so-called "birthday problem" is related to the probability that, given a set of randomly chosen people some pair of them will share the same birth date. Although it does not seem immediately intuitive, after examining 23 pairs of individuals, there is a 50-50 chance of two people in the room sharing a birthday. And after only comparing 57 pairs of individuals, there is a 99% chance of a match between *any* two individuals. With a list of 23 people to compare, there are $23 \times (23\text{-}1)/2 = 253$ possible pairwise comparisons. The first individual is compared to the other 22, the second individual to 21 people (excluding the first individual who was already compared), and so forth until everyone has been compared to everyone else.

When the question is changed to how many people match a specific birthday, then the numbers of individuals that need to be available changes. In order to reach a greater than 50% probability that another people will share a *specific* birthday, such as April 1, at least 253 individuals will need to be queried.

The birthday problem asks whether *any* of the people in a given group has a birthday matching *any* of the others—not one in particular. Note that in the birthday problem, neither of the two people is chosen in advance. When a specific DNA profile is developed from crime scene evidence, the question has changed. Typically with a DNA profile comparision, the question is not whether *any* profiles match (e.g., 23 in the birthday example or Figure 8.3d) but whether another specific profile exists (e.g., 253 in the birthday example or Figure 8.3b).

Source:
http://en.wikipedia.org/wiki/Birthday_problem

the FBI Laboratory provide a copy of the U.S. national DNA database to them (and presumably other researchers) to study the STR allele frequencies contained within the database and to look at the occurrence of partial matches (Krane et al. 2009). The FBI has so far denied this request citing concerns about genetic privacy (Geddes 2010). According to the FBI, federal law prohibits sharing DNA profiles from the U.S. national database with noncriminal justice agencies.

While it would be interesting to examine allele frequencies and perform other potential research on the data within the U.S. national DNA database, it would be challenging to draw meaningful correlations to other sets of population-specific allele frequencies. As already previously noted in this chapter, twins and sample duplicates exist in the database. In addition, samples are not identified or categorized by ethnicity, preventing the ability to determine true Caucasian or African American allele frequencies. Perhaps in the future, a large data set with millions of DNA profiles can be examined, but it is doubtful that this information will dramatically change our understanding of DNA results or match statistics.

WHEN THERE ARE NO HITS AFTER A DATABASE SEARCH

If a perpetrator of a particular crime has not been arrested or convicted of another crime (and had a DNA sample previously collected and uploaded to a DNA database), then no amount of searching the database with a particular crime scene profile will help solve the case. Searching the database again at a future date may be helpful if the perpetrator has been included in the meantime for another crime.

Three approaches have been taken when no match was found following a DNA database search: (1) the issuance of a so-called "John Doe warrant" based on the evidence profile, (2) conducting a DNA dragnet or mass intelligence screen by collecting samples from "volunteers" in a focused area, or (3) expanding the effective size of the database by conducting familial searches where a partial match between the evidence DNA profile and an offender may help focus the investigation on a relative of the offender in the database. The challenges of each approach are discussed below. Efforts are also being made to predict phenotypic information (see Chapter 12) including appearance and age of an individual based on DNA information from the evidentiary sample—something that cannot be done with the STR loci in use for human identity testing.

With particularly heinous crimes, additional efforts or strategies are sometimes considered to help solve the case. These approaches can involve casting a wider net through collecting additional samples (DNA dragnets) or using currently available DNA profiles with expanded searching for possible relatives (familial searching).

John Doe Warrants

Many states have *statutes of limitations* meaning that, after a certain period of time, a crime cannot be prosecuted. In order to stop the clock on the statute of limitations for commencing a criminal case, "John Doe warrants" have been used in a number of cases (Gahn 2000). This action can then extend the timeframe for possibly solving a case. If DNA evidence exists from a crime scene yet no suspect has been located to be charged with the crime, a John Doe warrant may be issued based solely on the assailant's genetic code.

The capabilities of forensic DNA testing have generated new legal issues for prosecutors. The sensitivity of the polymerase chain reaction enables DNA profiles to be obtained from previously intractable evidence. Furthermore, the existence of DNA databases now permits matches between perpetrators of crimes spanning jurisdictions and cold hits on unsolved crimes many years after they occurred.

In September 1999, Norman Gahn, Assistant District Attorney from Milwaukee County, Wisconsin, filed the first warrant for the arrest of "John Doe," an unknown male who could be identified by his 13 locus STR profile (Gahn 2000). This approach has been successful in stopping the ticking clock of a crime's statute of limitations, making it possible to prosecute the crime when the assailant is identified through a DNA database cold hit in the future. Several of these John Doe's have been subsequently identified with DNA database cold hits and successfully prosecuted for the crimes they committed.

Wisconsin law governing the statute of limitations was amended in September 2001 to provide for the use of DNA profiles from individuals unknown to the prosecution at the time the warrant for arrest is issued. The new legislation creates an exception to the time

limits for prosecuting sexual assault crimes if the state has DNA evidence related to the crime. John Doe warrants have also been issued in other states.

DNA Dragnets Through Mass Screens

In the past, there have been instances when a DNA dragnet was instituted if a DNA database search did not link any offenders on the database to a particular crime scene sample(s). Prior to the availability of national DNA databases, these DNA mass screens were more common.

The first use of forensic DNA testing involved a genetic dragnet of over 4,000 adult males in the Narborough, England area in 1986 and 1987. Samples that failed to be excluded from the crime scene sample with traditional blood typing were subjected to "DNA fingerprinting" or multi-locus RFLP testing. Colin Pitchfork was eventually apprehended based on this mass DNA intelligence screen and additional police work. However, challenges have been raised to the cost-effectiveness of this mass screening approach. For example, Colin Pitchfork was apparently the 4,583rd male tested for the mass screen described in Joseph Wambaugh's *The Blooding*.

DNA intelligence or mass screens to aid identification of a perpetrator and exclusion of innocent individuals in no-suspect cases have been successfully used many times by the UK's Forensic Science Service and other law enforcement agencies. In these mass screens, the police ask individuals within a predetermined group (e.g., males 18 to 35 years old within a defined location) to voluntarily provide DNA samples in an effort to identify the perpetrator of a crime or a series of crimes that have been linked by DNA evidence.

The largest mass screen conducted to date by the UK's Forensic Science Service was in conjunction with the investigation of the murder of Louise Smith, whose body was found near Chipping Sodbury, England in 1996. Over 4,500 samples from local volunteers were analyzed at an expense of over one million pounds. Eventually police realized that one of the potential suspects had since moved to South Africa. He was tracked down and his DNA sample taken, which was found to match a crime scene STR profile recovered from the scene. David Frost is now serving time for the crime he committed.

Of course, this type of effort and expense is not expended in every case but it has proven useful in some situations. However, collecting samples from every individual fitting a particular description or living in a particular geographic region is not always well received by the general public. Questions about genetic privacy and civil liberties are often raised, particularly in the United States, when mass screens are initiated. Within the United States, these types of DNA dragnets raise Constitutional concerns such as violation of Fourth Amendment rights that protect an individual's privacy until evidence is produced and a warrant obtained that would compel an invasive search.

In probably the largest genetic screening effort ever conducted within the United States, over 2100 individuals in the Dade County (Miami), Florida area were typed with AmpliType PM and HLA-DQA1 during the fall of 1994 (Kahn et al. 1996). Since six homicide victims had been prostitutes who were killed within a three mile radius between September 1994 and January 1995, tested individuals were primarily selected based on a prior arrest record for soliciting a prostitute. During the course of the investigation, three of the 2100 individuals tested matched with the PM + DQA1 screening assay, but

subsequently were excluded by follow-up RFLP analysis, which has a higher power of discrimination (see Butler 2010, *Fundamentals*, Chapter 3). In the end, the serial rapist/murderer was caught when a potential victim freed herself and police were called. Thus, while the perpetrator's DNA was found to match DNA evidence from the previous six crime scenes (and this information was used in his prosecution), the mass screen was not responsible for his apprehension.

In April 2004, a DNA dragnet was conducted in Charlottesville, Virginia to try to stop a rapist who had attacked at least six women between 1997 and 2003. Community concerns that black men were being targeted led police in Charlottesville to eventually suspend the mass screen after only collecting and analyzing samples from about 200 men. The perception of being "guilty" before proven "innocent" through DNA testing is an issue often raised by critics of DNA dragnets—particularly in the United States where the criminal justice system attempts to treat individuals as "innocent until proven guilty."

A September 2004 report that surveyed DNA mass screens conducted in the United States over the previous two decades found these efforts to be "extremely unproductive" in identifying the true perpetrator (Walker 2004). This report recommended that police not conduct DNA mass screens based on general descriptions of criminal suspects. On the other hand, a study presented to the European Network of Forensic Science Institutes (ENFSI) DNA Working Group in April 2006 found that 315 out of 439 mass screens conducted in Europe were successful in identifying the perpetrator (Wenzel 2007). The 72% success rate in Europe illustrates that DNA intelligence-led screens can be helpful in solving crimes when other efforts have failed—although the approach may not be well accepted by the public in some areas. *It is hoped that the proper balance can be found in the future to fully utilize the power of DNA testing and yet preserve the privacy and civil liberties of innocent citizens.*

Partial Matches and Familial Searching

In a May 2006 *Science* article entitled "Finding criminals through DNA of their relatives," the authors propose that if a crime stain does not match anyone in the offender database that there is a chance that a relative might be in the database (Bieber et al. 2006). Since relatives will have similar DNA profiles, loosening the search stringency to permit partial matches rather than full high-stringency matches (where every allele in an STR profile must match) may return a list of results that could include a brother or other close relative. This list of potential relatives could be narrowed through further testing with Y-chromosome markers (for males only), which would require all of the potential relatives plus the crime scene sample to be examined with the additional genetic markers (Myers et al. 2010). In theory, with this approach, the use of offender profiles already on the DNA database enables close relatives of offenders to potentially be included in searches performed against DNA profiles coming from unsolved crime scenes.

The UK pioneered this partial matching technique, better known as "familial searching," and has used it to solve a number of cases—but not without controversy.

NDIS defines a "partial match" as a moderate stringency candidate match between two profiles having at each locus *all of the alleles* of one sample represented in the other sample. It is the result of a routine search at moderate stringency where the forensic scientist, when evaluating whether a candidate match is a viable match and should be processed through

to confirmation, discovers that the candidate offender is, in fact, excluded as the possible perpetrator of the crime. However, because of a similarity in alleles between the target and candidate profiles, the analyst believes that a close biological relative of the offender could possibly be implicated in the crime. On the other hand, a familial search is a deliberate search looking for relatives.

A successful California familial search announced in July 2010 captured the attention of the world and brought a serial offender known as the Grim Sleeper to justice (D.N.A. Box 8.5, Miller 2010). This case certainly demonstrates the potential and power of familial searching when police have no other possible leads and the perpetrator poses a serious threat to public safety. However, the technique has technical pitfalls and privacy concerns that are discussed in Appendix 2.

D.N.A. BOX 8.5

THE "GRIM SLEEPER" CASE: A FAMILIAL SEARCH SUCCESS STORY



On the morning of July 7, 2010, officers from the Los Angeles Police Department (LAPD) waited quietly and impatiently outside the home of Lonnie Franklin, a 57-year-old mechanic. When he finally stepped outside, he was immediately surrounded and arrested on serial murder charges. His arrest brought closure to one of the most prolific perpetrators in L.A. history, but at the same time, opened the door nationwide to questions about the methods used to identify him. The police showed up on Mr. Franklin's doorstep because of DNA collected initially not from him but from his son. It was the typing of this DNA sample, combined with a specialized DNA search technique known as "familial searching" and the persistence of investigators who used the cutting edge capabilities of DNA testing that led to his arrest.

The investigation into a serial killer began with a DNA match—but one without a suspect. In June 2007, an LAPD task force was established to locate an active serial killer because of a case-to-case DNA hit identified by the LAPD Crime Lab a month earlier. The lab had found that DNA evidence recovered from a victim located in a South Los Angeles alley in January 2007 matched a previously analyzed 2003 case. Eventually, DNA matches would be made to a series of cases from the 1980s.

From 1985 through 1988, a killer had stalked the streets of Los Angeles murdering seven women and one man, who ranged in age from 18 to 36. A ninth victim was raped but survived in November 1988. A number of the victims were troubled women who

D.N.A. BOX 8.5 (*cont'd*)

were vulnerable and were out on the streets at night as prostitutes. These crimes were initially connected with forensic evidence involving bullet ballistics coming from a 0.25 caliber gun. However, no leads existed as to the person behind the crimes. Three more murders would be committed in March 2002, July 2003, and January 2007 (the last two were the first two connected by DNA testing).

In reviewing the cases, the LAPD found a 13-year gap in detected victims—leading an *LA Weekly* reporter in August 2008 to nickname the serial killer the "Grim Sleeper." The name stuck—and a media campaign was launched with billboards and a $500,000 reward for information leading to the killer's arrest. A common DNA profile, believed to be that of the perpetrator, was eventually developed from six of the 12 victims connected to this serial killer. Unfortunately, searches of state and national DNA databases yielded no result. The killer had somehow managed to keep out of reach from the long arm of the law for almost 25 years.

Meanwhile, the California Department of Justice (CA DOJ) DNA Laboratory (in Richmond, CA, northeast of San Francisco), which manages the offender DNA database in California, was developing software based on kinship analysis and performing validation studies on familial searching, a somewhat controversial technique that had seen some success in England. By reducing the stringency of a DNA search, the UK's Forensic Science Service (FSS) had shown that DNA profiles from unsolved crimes could be linked to relatives of the perpetrators who were present in their national DNA database. Used in 158 UK cases from 2003 to 2008, the FSS was able to close some 18 cases because of familial searching. In April 2008, the CA DOJ DNA laboratory, with the approval of the Office of the Attorney General, announced that it would perform familial searches of the state DNA database in cases of serious crime when all other methods had been exhausted. By October 2008, studies were complete and validated software was in place. They now had the tool to effectively widen their search net.

With the public and the victims' families clamoring for capture of the killer and closure to the serial murder cases, the LAPD sent a formal request to CA DOJ to perform a familial search of the state DNA database. In October 2008, this specialized search against the California state DNA database, which at that time contained over 1.1 million convicted offender profiles (1,102,659 to be exact), was performed looking for a possible relative of the "Grim Sleeper" evidence profile. No likely relatives were identified by this first search—leaving the LAPD empty-handed once again. "It was very discouraging," says Detective Dennis Kilcoyne, who led the LAPD task force, "[by June of 2010] we had spent three years essentially beating our heads against the wall and expending a lot of resources without any result."

Over the course of the 18 months following the initial search hundreds of thousands of new DNA profiles poured into the California state DNA database. In the spring of 2010 (after weekly direct searches of the database failed to locate a match), another formal request was made by the LAPD task force to search the offender database again using the familial searching approach. On June 30, 2010, Detective Kilcoyne received

D.N.A. BOX 8.5 (*cont'd*)

a call from Jill Spriggs, chief of the CA DOJ's Bureau of Forensic Services, asking him to quietly arrange a meeting later that week with the chief of police and the district attorney. Friday, July 2, Chief Spriggs and CA DOJ's Bureau of Investigation and Intelligence (BII) Chief Craig Buehler flew to LA and shared the results of their familial search.

Through a video conference with the CA DOJ scientists in Richmond, the LAPD learned that a two-part scientific screen, followed by careful investigative confirmation, had been used to locate a potential relative of the perpetrator. First, a ranked list of ≈150 individuals with the highest likelihood ratios, which indicated possible familial relationships with the Grim Sleeper DNA profile, had been developed from the state database, at that time containing 1,322,998 convicted offender profiles. Then these DNA samples were compared using Y-chromosome testing to eliminate all non-paternally related offenders who shared similar autosomal STR characteristics by chance. Most of these individuals had already been eliminated by Y-STR testing during the first familial search. Only a single individual remained on the list after the Y-STR testing—a man by the name of Christopher Franklin, who had alleles matching the Grim Sleeper DNA profile at all 15 tested autosomal STR loci.

Christopher Franklin's DNA profile had been added to the state database in early 2009 due to his conviction on a felony weapons possession charge. Thus, his DNA profile was not present during the initial October 2008 search. Further investigative footwork was performed by CA DOJ BII agents through public records searches to learn more about potential family members who may have lived in the LA area during the times of the crimes. All of this investigative work was performed *without talking to any member of the general public* and prior to giving the name of a potential suspect to the LAPD detectives. The 2010 familial searching results pointed to the biological father of Christopher Franklin, who lived in the vicinity of the Grim Sleeper crimes.

Within a few hours, Lonnie David Franklin, Jr. had over 20 detectives shadowing his every move and waiting for him to discard some biological sample in a public place so that the police would have a direct DNA reference sample to confirm their suspicions that he was the Grim Sleeper serial killer. It is important to note that Mr. Franklin had not previously been on the police radar as part of this investigation—and probably would not have been without the familial DNA association. Moreover, the police quickly discovered that he was not a litter bug and did not dispose of any useful items, such as cigarette butts or soda cans, within sight of their surveillance.

Finally, on Monday afternoon, July 5, Mr. Franklin went to a pizza restaurant to eat. An undercover officer became his waiter—and according to Detective Kilcoyne, Mr. Franklin probably received the best service he ever had in a restaurant. His dirty napkins were replaced with clean ones and his fork was removed if it was dropped. At the end of his meal, a piece of leftover pizza crust was also collected. In total, eight items were submitted to the LAPD DNA Lab for testing—and analysts worked quickly to generate results on the submitted items.

D.N.A. BOX 8.5 (*cont'd*)

On Wednesday morning, July 7, about 30 hours after submitting the samples, the LAPD Task Force received the laboratory reports. DNA on the napkin and the pizza crust eaten by Mr. Franklin was a direct match to the Grim Sleeper evidentiary DNA profile. With this DNA confirmation, the LAPD completed their preparations to perform the arrest. When Mr. Franklin came out of his house around 9:20 a.m. to do something in his yard, he was greeted by police officers with an arrest warrant. To further verify the results, the police immediately collected a buccal swab from Mr. Franklin to confirm the DNA match. Then, they proceeded to conduct a three-day search of his home to collect evidence. Lonnie David Franklin, Jr. was subsequently charged with 10 murders and is awaiting trial as of early 2011.

In many ways, this was a perfect success story for familial searching using DNA databases. A series of heinous crimes could not be solved any other way despite years of investigative effort. An initial search before the relative was in the database led to no further action. Y-STR testing for confirmation purposes on the second search yielded only one profile. Through public record searches—and without disturbing any potential family members—investigators learned that the individual in the database had a father who was the appropriate age and in the appropriate location to have committed the crimes.

In an opinion article a few days after Mr. Franklin was arrested, the *New York Times* praised the careful and conscientious efforts of the scientists and police investigators involved in this case. Even the American Civil Liberties Union (ACLU), typically a harsh critic of DNA databases, was complementary of the careful investigation in this case while protecting individual privacy and civil rights.

Detective Dennis Kilcoyne headed the task force, which was established in June 2007 to identify this serial killer who operated for at least 25 years without detection. Detective Kilcoyne believes that this case would not have been solved without the power of DNA testing and particularly familial searching. Both the CA DOJ scientists and LAPD investigators were aware that they were doing cutting edge work and were careful to preserve privacy at each stage of the investigation. In particular, a detailed protocol established by the CA DOJ laboratory was followed to protect privacy, while maximizing the reach of DNA testing, thus establishing an effective model for future familial searches. While familial searching currently has a success rate of 10% in California (as the nine previous searches failed to provide any leads), to a dedicated detective like Dennis Kilcoyne, this powerful technique provided closure on a critical case and brought a heinous killer to justice.

Sources (additional references are available at end of Chapter 8 and Appendix 2):

ACLU (2010). "Grim Sleeper" case doesn't justify expanding the reach of DNA databases, Available at http://www.aclu-sc.org/news_stories/view/102863/.

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CONCERNS OF DNA DATABASE CRITICS

The rapid expansion of DNA databases and their use to solve crimes has many privacy advocates concerned (Greeley et al. 2006, Krimsky & Simoncelli 2010). Within the United States, numerous legal challenges have tried unsuccessfully to claim that DNA collection violates Constitutional rights of American citizens. In any society, there are tradeoffs between public safety and individual privacy. In order to help readers better understand the concerns of DNA database critics, a few of the most common concerns are discussed below.

Constitutionality of Collection

During the time period of 1990 to 2010 in the United States, there have been over 200 reported cases brought to court relating to the constitutionality of federal and state DNA databank sample collection and DNA database searches (Dawn Herkenham, personal communication). Time after time the constitutionality of criminal DNA databases and databanks has been upheld.

Likewise, there is not any expectation of privacy for genetic material discarded at a crime scene (e.g., from a cigarette butt or from a semen stain). However, the more recent collection of DNA samples from arrestees has raised concerns that Fourth Amendment Constitutional rights against warrantless, suspicionless searches are being violated (Krimsky & Simoncelli 2010).

Arrestee Testing

As can be seen from Table 8.7, there are an increasing number of states moving to arrestee testing so that DNA profiles can be entered earlier in the criminal justice process. Currently, NDIS holds these records in a separate index from convicted offenders (see Table 8.2). Within the United States, as of late 2010, 25 states require individuals arrested for felony offenses to provide a DNA sample for inclusion in the state DNA database. Many countries around the world are also moving towards arrestee DNA testing.

If an arrested individual is not convicted, many of these laws require that the DNA profile be removed from the database and the DNA sample be destroyed. However, most laws call for the acquitted individual to request the profile expungement and sample destruction.

As the first state with experience implementing an arrestee index at SDIS, Virginia found that database hits for unsolved crimes could be made earlier and thus potentially prevent future crimes by the same individuals (Ferrara & Li 2004). In their first eight years of operation (2003 to 2010), Virginia observed over 625 hits to the arrestee index (see http://www.dfs.virginia.gov/statistics/index.cfm). Arrestee sample collection does result in connecting crimes through DNA, but more research would be helpful to see the real impact of making this connection earlier in the criminal justice process (i.e., at arrest instead of waiting for conviction).

Racial Distribution of the U.S. Database

A common argument of DNA database critics (e.g., see Chapter 15 in Krimsky & Simoncelli 2010) is that there is a higher representation of minorities (e.g., African

Americans and Hispanics) on the U.S. as well as other national DNA databases. Since LDIS, SDIS, and NDIS do not keep records as to the ethnicity or racial background of DNA profiles, this allegation cannot be confirmed. However, U.S. prison population statistics are available from which it may be possible to infer the potential racial composition of the NDIS offender index, which is composed of convicted offenders. This correlation of course is based on the assumption that all offender samples are being collected and placed into the DNA database.

Table 8.10 shows that a much higher number of African American (black) and Hispanic individuals are in U.S. prisons relative to the number of Caucasians (white). For example, there are 6.6 times as many African American versus Caucasian males ages 30 to 34 in prison and 2.2 times as many Hispanic versus Caucasian males of the same age range. If these prison population numbers correlate to the racial makeup of NDIS (or SDIS or LDIS), then as critics claim there is a higher number of non-Caucasian DNA profiles in the database.

However, it is important to note that the argument about minorities being overrepresented on the DNA database has more to do with the criminal justice system as a whole and not the DNA database itself. The DNA samples that are put in the database are supposed to be collected at the local, state, or federal level from convicted offenders or those arrested for a legislated qualifying offense. *There is no pre-selection of certain population groups in terms of the DNA collection*. Whether or not individuals have their DNA collected has to do with whether or not they are convicted of or arrested for a crime that qualified them for DNA submission.

TABLE 8.10 Estimated Number of Inmates Held in Custody in State or Federal Prison, or in Local Jails per 100,000 U.S. Residents, by Sex, Race, and Age as of June 30, 2009.

	Male			Female		
Age	**White**	**Black**	**Hispanic**	**White**	**Black**	**Hispanic**
18–19	776	4403	1838	70	210	140
20–24	1389	8889	3937	194	595	329
25–29	1569	10,501	3954	222	733	314
30–34	1673	10,995	3650	260	896	302
35–39	1587	10,068	3090	263	895	300
40–44	1475	8668	2735	214	730	248
45–49	972	6387	2327	115	405	191
50–54	568	3914	1583	63	155	132
55–59	383	2203	1159	25	60	86
60–64	227	1134	758	9	49	36
65 or older	87	454	243	3	5	9

Source: http://bjs.ojp.usdoj.gov/content/pub/pdf/pim09st.pdf (Table 19 data).

A Universal Population-Wide Database?

As DNA database expansion has occurred, there have been calls by some for a universal database in which profiles of everyone in the population would be included on the database (Kaye & Smith 2003). The argument for doing so is typically two-fold. First, if more DNA profiles are in the database, then the advocates of this approach suggest that there will be an increase in the hit rate. This assumes that a probative biological sample will be left at most crime scenes from which a useful DNA profile can be obtained.

The second common argument for creating a universal database stems from the concern raised in the previous section, namely that there is a disproportionate level of minorities' samples in the DNA database. The reasoning of the universal database advocates is that if everyone is put in the database then there would be no racial inequality in terms of sampling.

A number of technical as well as ethical issues would exist with efforts to create a universal database. First, currently it would be expensive to generate a population-wide DNA database and would overwhelm a system that already has DNA backlogs (see Table 8.9). Moreover, if a DNA database contains a large number of innocent people who never commit crimes, the system would be bogged down by the expense of testing these individuals on the front end. Database searches would be clogged on the back end by having to sift through all of the extraneous data. Increasing the database size that dramatically (e.g., 300 million within the United States) would require more powerful search algorithms and much greater computer capacity. Additional loci would need to be tested to increase the profile random match probabilities to avoid adventitious matches (see D.N.A. Box 8.3).

The collection of samples would create both logistical and ethical issues. Among adults and juvenilles of crime-committing age, how do we go about collecting DNA from innocent people? What if someone is not willing to volunteer their sample for DNA testing? Would some form of coercion be used to collect the sample? Measures would have to be in place to avoid duplication of sample collection and processing if people move around. In other words, collection would not be a trivial task.

Would DNA be collected from children, for example, at birth or at a very young age? The benefit of collecting the children's DNA profiles at birth would not be seen until many years into the future as those who choose to commit crimes would not likely start until their late teenage years or older (and hopefully they would never start committing crimes). As a society, are we willing to collect DNA from innocent children as an investment for the future?

Predicting where we will be 20 years into the future is unrealistic in terms of the technology and genetic markers that will be in use. We might be typing genetic markers that may be obsolete 20 years from now. For the DNA profiles to be of value once the children grow old enough to commit crimes, the samples would have to be retained and tested or re-tested with new genetic markers of the future.

Would knowing that their DNA profile is on a database serve as a deterrent to bad behavior for people? Perhaps for some individuals. Unfortunately, as the SFPD study showed, serious repeat offenders keep offending even on probation (Gabriel et al. 2010)! DNA may be a deterrent to some criminals but certainly not to all potential offenders (see Prainsack & Kitzberger 2009).

Summary and Final Thoughts

As DNA databases continue to demonstrate their success in aiding law enforcement investigations, they will expand in size. New laws that effectively widen the search net by including offenders from an expanded list of crimes will bring in more DNA samples. The DNA testing community must cope with this growth and overcome sample backlogs. By the end of the first decade of the 21st century, the U.S. national DNA database is growing at a rate of more than one million new samples each year. Growth gives rise to operational challenges. Faster search algorithms are needed, additional core loci will need to be included to avoid adventitious matches, and safeguards on the application and use of the database need to be maintained.

DNA databases have shown their value in benefiting the criminal justice system. Database hits have linked serial crimes and located perpetrators who left biological evidence at crime scenes years earlier. Post-conviction exonerations of innocent people who have spent years behind bars have been made possible through DNA database connections to the true perpetrator. Any tool that is this powerful will be criticized and needs to be used responsibly in order to balance public safety with individual liberties.

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